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A semi-analytical fractional order neural network framework for two-dimensional time fractional reaction-diffusion problems

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A B S T R A C T

In this research, we provide a convergence of the Adomian decomposition based neural network method for time fractional reaction diffusion initial boundary value problems in 2D, where the fractional time derivative is considered in the Caputo sense. We propose an idea of a proper combination of Adomian decomposition method (ADM) and Physics-informed neural network (PINN), after semi-discretization of the time fractional derivative. PINN uses the ADM based partial sum inside the loss function before generating the appropriate ADM-PINN approximation. In addition, we produce the required sufficient conditions on the given data under which the usual Adomian decomposition method converges in a bounded domain. Furthermore, we analyze the error bound of the proposed method for the time fractional model and demonstrate the convergence. Several numerical examples are produced to show the effectiveness of the present ADM-PINN approach. It is observed that the proposed method is highly effective for several two-dimensional time fractional problems including the Schrodinger equation for convergent approximation of the computed solution.

1. Introduction

Deep learning algorithms are prominent tools in the field of scientific computing, as their universal approximation property enables the representation of complex solutions arising from differential equations (LeCun et al., 2015; Liang & Srikant, 2016). In recent years, deep neural networks have been widely adopted for solving a broad range of problems, with notable success in the field of computer vision and image processing, where they pose significant challenges to traditional numerical methods (Drogoul, 2014). In the field of engineering, control systems mainly face two major issues, namely stochastic abrupt changes in system operations and complex nonlinear dynamics. To effectively manage the sequential growth rates, event-triggered conditions, and thresholds, the authors of Shanshan et al. (2025) created a novel memory-based adaptive event-triggered mechanism. The information at the triggering instant has been taken into consideration using the Lyapunov-Krasovskii functional. Similarly, the study (Li et al., 2025) investigates an event-triggered control technique for stochastic multi-agent systems to complete consensus tracking with less energy and communication. They consider the influence of noise (follows Brownian motion) in the control environment and modify the communication topology according to a Markovian switching process. Furthermore, the algorithm updates the controller only when the tracking error exceeds a predetermined threshold. Finally, they validate the effectiveness of the methods through numerical experiments. Lagaris et al. (1998) employed an artificial neural network (ANN) to solve differential equation and a system of differential equation, by constructing the trial solution that exactly satisfies the initial and boundary conditions. The numerical simulations demonstrated on the several benchmark problems and outperformed the finite element approach based on error and computational efficiency. Moreover, the meshless property of deep learning algorithms puts numerical techniques on the back foot for solving linear/nonlinear PDEs (Liu et al., 2023; Patawari & Das, 2025), higher-dimensional semilinear parabolic PDEs (Ghosh et al., 2026; Han & Jentzen, 2017), and singularly perturbed problems (Das & Mehrmann, 2016; Kumar et al., 2024; Patawari et al., 2025; Saini et al., 2024; Shakti et al., 2022). Raissi et al. (2019) introduced Physics-Informed Neural Networks (PINNs), a deep learning framework that used the physical phenomena of differential equations

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to solve the forward and inverse problems, even in the presence of scattered and noisy data. In the sequence of theoretical justification, the authors of Shin et al. (2020) provided the first rigorous error bound and theoretical convergence discussed for 2nd-order linear elliptic and parabolic PDEs. They have observed that as the width of partition sizes decreases, the PINN approximation converges to the exact solution of PDEs, using the Schauder fixed point approach and the maximum principal. Furthermore, the authors of Mishra and Molinaro (2022) used an abstract formalism and quadrature rule to provide the generalization error of PINN for linear and quasilinear incompressible Euler equations, which depend on the training points. This work is also extended to obtain the generalization error for the Navier-Stokes equations mentioned in De Ryck et al. (2024).

The traditional PINN uses automatic differentiation, backpropagation, and an appropriate optimizer to solve the differential equations. Automatic differentiation employs the standard classical chain rule to compute the integer-order differential operator. The standard chain rule creates a research gap between integer-order and fractional-order differential equations approximated through PINN. The authors of Pang et al. (2019) consider this issue and developed fractional PINN, which is an extension of standard PINNs known as fPINN. The fractional PINN involves automatic differentiation for integer-order operators and numerical schemes for fractional-order operators. They have shown the convergence of PINN through space-time fractional advection-diffusion equations and multidimensional forward and inverse problems, and provided a comparison analysis with the finite difference approach. In the present paper, we modify the PINN technique to solve the time fractional reaction-diffusion equation (Cao et al., 2025), which have gained attention in the last decade because of the failure of the automatic differentiation. Applications of the fractional-order reaction-diffusion model occur in several places, like; in the modeling of activator-inhibitor dynamics with anomalous diffusion in inhomogeneous media (Henry & Wearne, 2006), where the reaction term and the diffusion term account for the growth and migration of the species, respectively. In the similar fashion, the authors of Sahoo et al. (2025) tackle geophysical Korteweg-de Vries equation with time fractional derivative that arises from uncertainty in the climate changes or the dynamics of air and water waves introduce uncertainty derivative. The authors of Ren et al. (2023) extend the fractional physics-informed neural network to tackle oscillations inside boundary layers for singularly perturbed problems computationally. The authors of Shekarpaz et al. (2024) recently developed a splitting-based PINN algorithm, by extending the PINN and fPINN approaches to efficiently predict the oscillations arising in fractional and integer order problems. We can also find some recent papers on fractional PINNs (Mehta et al., 2019; Zou et al., 2024), a survey on computational aspects (Das et al., 2026). In the field of scientific computing, the fractional-order reaction-diffusion model is a growing area in the current era. In today's literature, solution approximations on space or time adaptive meshes are frequent (Choudhary et al., 2024; Das , 2019; Das & Vigo-Aguiar, 2019; Kumar et al., 2024; Santra et al., 2023) when there is a layered phenomenon in space or time. Other approaches include Operator splitting (Baeumera et al., 2008) for two-dimensional models, and fractional Fourier transform using Fourier spectral analysis (Pindza & Owolabi, 2016). However, the extension of PINNs for problems with bounded or unbounded domains which leads to faster convergence is still open. In this work, we combine a semi-analytical Adomian decomposition method with PINN and provided its analysis for time-fractional reaction-diffusion problems in a bounded domain. We believe that our provided approach will help further research communities working on either PINN or on-time fractional problems. Now we explain the preferred decomposition method.

The Adomian decomposition Method (ADM) utilizes a partial sum-based series approximation to generate a convergent solution. Here, we provide the convergence analysis of the approximation of reaction-diffusion models based on this decomposition, which was initially introduced by Adomian (Adomian, 1988). The combination of ADM and PINN can lead to a better result than PINN if one can identify a minimum number of specific terms, and it is also superior to the ADM solution. However, there is no hard and fast rule to find this number if the initial ADM approximations are able to detect the time singularity (Sabawi et al., 2025; Santra et al., 2025). Hence, we propose a combination of PINNs with the ADM, which provides a convergent and better approximation if the initial guess from the semi-analytic approximation is provided. In the aspect of ADM literature, it is observed to be effective for several problems, like the Fokker-Planck equation (Dehghan et al., 2007), Duffing equation and Van der Pol oscillators (Adomian, 1988), time fractional Burger's equation (Li & Wang, 2009), nonlinear fractional equation (Rach et al., 2013), Gurtin-Pipkin integro-differential equations (Anukiruthika et al., 2025), etc., irrespective of linear and nonlinear cases. Moreover, the authors of (Ahmad et al., 2025) used a mesh-less collocation method based on Gaussian radial basis functions to solve the two asset time-fractional Black-Scholes models for American and digital options, where Caputo derivative is used to handle the time-fractional component. In particular, this approach can also be treated as an alternative approach over the numerical discretization, as the decomposition may even lead to the exact solution of the model. The ADM is a semi-analytic method like homotopy perturbation (Zhang et al., 2017) and iterative methods (Al-Jawary et al., 2018), which reduces the original problem into a convergent sequence of simpler subproblems. The convergence of the approximate solutions based on ADM was first considered by Cherruault (Cherruault, 1989), Cherruault and Mavaungou (Mavoungou & Cherruault, 1992). However, they have mainly considered the initial value problems for PDEs in the unbounded space domain, and the extension of this method for initial boundary value problems has not yet been explored. The effectiveness of semi-analytical ADM for fractional order nonlinear Heat and Wave equations is also observed in Ray and Bera (2006) for initial value problems.

The main motivation of the present work is to provide a novel framework for the ADM-PINN algorithm. We have discussed a vast amount of ADM literature, and it is observed that this method is effective for most of the initial value problems when the space domain is unbounded. Here, we analyze the theoretical and numerical convergence of the ADM for initial boundary value problems for time fractional reaction-diffusion models in bounded domains. We prove theoretical convergence and provide sufficient conditions on the initial data under which one can solve the initial boundary value problems on the bounded domain with the usual decomposition procedure defined in Eq. (6). A key limitation of the ADM is based on the slow convergence that may require a large number of series terms for a reasonable error, which may be hard to interpret practically. To reduce the number of series terms with a reasonable error, we introduce a neural network framework along with the ADM data. Automatic differentiation is an important aspect in the neural network framework to compute the differential operator. Automatic differentiation uses the classical chain rule that is valid for an integer-order differential operator. Therefore, we need proper care for the fractional operator, and here, we use the L1 scheme for approximating it, and convergent ADM data is used in the loss function that provides optimal parameters to the neural network framework. The main novelty in the neural network is to introduce an additional loss inside the traditional loss, which includes the prior information of the solution from the ADM algorithm-generated data, and provide the optimal choices to the neural network parameters. Finally, we prove the error bound and theoretical convergence of the ADM-PINN algorithm. The outlines of the paper are as follows. In Section 2, we define a two-dimensional fractional reaction-diffusion model on a bounded domain. In Section 3, we introduce the ADM algorithm procedure and the semidiscrete version of the present problem before introducing PINN for time fractional semi-discrete problems. In Section 4, we provide the convergence analysis of the decomposition method in a bounded domain and provide sufficient data under which the convergence of the decomposition method is guaranteed. In Section 5, we provide the ADM-PINN combination algorithm and provide the convergence analysis of the generalization error based on PINN. In Section 6, we conclude the effectiveness of the PINN over the ADM solution using numerical examples. Finally, we conclude the benefit of the present algorithm in Section 7.

Notations: For a domain $\Omega_{\mathbf{x}}$, we define its closure and boundary by $\partial\Omega_{\mathbf{x}}$ and $\overline{\Omega}_{\mathbf{x}}$ respectively. The supremum norm of a function $f(\mathbf{x})$ is defined as $\|f(\mathbf{x})\|_\infty = \max_{\mathbf{x}\in\Omega_{\mathbf{x}}} |f(\mathbf{x})|$ in a domain $\Omega_{\mathbf{x}}$. For an integer $m > 0$, $H^m(\Omega_{\mathbf{x}})$ is a Hilbert space. In particular, $H^m(\Omega_{\mathbf{x}}) \equiv \{v \in L^2(\Omega_{\mathbf{x}}) : D^l{}_{\mathbf{x}} v \in L^2(\Omega_{\mathbf{x}}),\ 0 \le \ell \le m\}$ where $D^\ell{}_{\mathbf{x}} u = \frac{\partial^\ell u}{\partial^{\ell_1} x \partial^{\ell_2} y}$ where $\ell = \ell_1 + \ell_2$ is ℓth order weak derivative of u and the corresponding norm is defined by $\|v\|_{H^m(\Omega_{\mathbf{x}})} = (\|v\|_{L^2(\Omega_{\mathbf{x}})} + \sum_{l=1}^m \|D^l{}_{\mathbf{x}} v\|_{L^2(\Omega_{\mathbf{x}})})^{1/2}$. In addition, we use $H_0^1(\Omega_{\mathbf{x}})$ which is defined by $H_0^1(\Omega_{\mathbf{x}}) = \{v \in H^1(\Omega_{\mathbf{x}}) : v = 0 \text{ on } \partial\Omega_{\mathbf{x}}\}$.

For a given Banach space $\mathcal{B}$ over space domain and for $m = 0, 1$, we define $H^m(0,T;\mathcal{B}) := \left\{u(t) \in \mathcal{B} \text{ a.e. for } t \in (0,T) \text{ and } \sum_{j=0}^m \int_0^T \left\|\frac{\partial^j u(t)}{\partial t^j}\right\|_{\mathcal{B}}^2 dt < \infty\right\}$ equipped with the norm $\|u\|_{H^m(0,T;\mathcal{B})} = \left(\sum_{j=0}^m \int_0^T \left\|\frac{\partial^j u(t)}{\partial t^j}\right\|_{\mathcal{B}}^2 dt\right)^{\frac{1}{2}}$. For a fractional order α, we define this space as $H^\alpha(0,T;\mathcal{B}) := \left\{u(t) \in \mathcal{B} \text{ for a.e. with } t \in (0,T) \text{ and } \int_0^T \left(\left\|u(t)\right\|_{\mathcal{B}}^2 + \|D_t^\alpha u(t)\|_{\mathcal{B}}^2\right) dt < \infty\right\}$ equipped with the norm $\|u\|_{H^\alpha(0,T;\mathcal{B})} = \left(\int_0^T \left(\left\|u(t)\right\|_{\mathcal{B}}^2 + \left\|D_t^\alpha u(t)\right\|_{\mathcal{B}}^2\right) dt\right)^{\frac{1}{2}}$. For simplicity, we denote the norm $\|.\|_{L^2(0,T;L^2(\Omega_{\mathbf{x}}))}$ throughout as $\|.\|$.

2. Reaction diffusion problem

In this paper, we consider the following two-dimensional time fractional reaction-diffusion initial and boundary value problem on $\Omega_{\mathbf{x}} \times [0,T]$ for a bounded domain $\Omega_{\mathbf{x}} \subset \mathbb{R}^2$:

$$\begin{cases} D_t^\alpha u(\mathbf{x},t) = a\,\Delta u(\mathbf{x},t) - b(\mathbf{x},t)\,u(\mathbf{x},t) + f(\mathbf{x},t), \\ \qquad 0 < \alpha < 1,\ (\mathbf{x},t) \in \Omega_{\mathbf{x}} \times (0,T], \\ u(\mathbf{x},0) = s(\mathbf{x}),\ \text{on } \overline{\Omega}_{\mathbf{x}}, \\ u(\mathbf{x},t) = 0,\ \text{on } \partial\Omega_{\mathbf{x}} \times [0,T]. \end{cases} \tag{1}$$

Here, the reaction term $b(\mathbf{x},t)$ considered as a sufficiently smooth positive function on $\overline{\Omega}_{\mathbf{x}}$ and a as a positive number. We are interested in approximating the solution $u(\mathbf{x},t)$ in $\overline{\Omega}_{\mathbf{x}} \times [0,T]$.

Here, the fractional derivative D_t^α of order $\alpha > 0$, is considered in the sense of Liouville-Caputo. This derivative for a function g, is defined as follows Podlubny (1999):

$$D_t^\alpha g(t) = \begin{cases} \dfrac{1}{\Gamma(1-\alpha)} \displaystyle\int_0^t (t-\tau)^{-\alpha} \frac{dg(\tau)}{d\tau} d\tau, & \text{if } 0 < \alpha < 1, \\ \dfrac{dg}{dt}, & \alpha = 1. \end{cases} \tag{2}$$

In addition, the Liouville Caputo fractional derivative can be related to the Riemann Liouville (R-L) fractional integral of order $\alpha > 0$ as follows

$$J_t^\alpha g(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-\tau)^{\alpha-1} g(\tau) d\tau, \quad t > 0, \tag{3}$$

where Γ defines the Gamma function, by the relation $D_t^\alpha g(t) = J_t^{1-\alpha} D_t^1 g(t)$. We can see Podlubny (1999), for a detailed discussion of the Caputo fractional derivative and its preliminaries.

The following result from [Theorem 4.1, Yamamoto et al. (2015)] provides the existence and uniqueness of the solution of (1) under sufficient regularity and compatibility on the data.

Theorem 1. *Consider $f \in L^2(0,T;L^2(\Omega_{\mathbf{x}}))$ in (1). Then, there exists a unique weak solution $u \in L^2(0,T;(H^2(\Omega_{\mathbf{x}}) \cap H_0^1(\Omega_{\mathbf{x}}))) \cap H^\alpha(0,T;L^2(\Omega_{\mathbf{x}}))$ of (1).*

3. ADM-PINN on fractional differential equations

We first explain the general procedure of ADM and describe how to combine ADM-PINN for solution approximations. Its convergence and error analysis will be considered in later sections. ADM approximates the analytical solution based on a special type of its series expansion. The partial sum of this series will be used for the initial guess for ADM-PINN-based approximations.

3.1. Operator decomposition method

Here, we discuss the ADM (Adomian, 1988) for a general problem. Let us write the problem (1) as follows:

$$L(u) + R(u) = f(\mathbf{x},t), \quad \text{for} \quad (\mathbf{x},t) \in \Omega_{\mathbf{x}} \times (0,T], \tag{4}$$

where L is considered as a linear fractional invertible operator and R defines the remainder of the linear operator. Let us decompose the solution in the following form

$$u(\mathbf{x},t) = \sum_{i=0}^\infty u_i(\mathbf{x},t), \tag{5}$$

where L^{-1} can be calculated by (3). Multiplying L^{-1} in (4) and using the above decomposition, we obtain the following recurrence relations of $u_i(\mathbf{x},t)$:

$$\begin{cases} u_0(\mathbf{x},t) = s(\mathbf{x}) + L^{-1} f(\mathbf{x},t), \\ u_j(\mathbf{x},t) = -L^{-1}\,(R\,u_{j-1})(\mathbf{x},t), \quad j \ge 1. \end{cases} \tag{6}$$

Let us define $\Phi_0(\mathbf{x},t) = 0$ and $\Phi_p(\mathbf{x},t)$ is defined as follows

$$\Phi_p(\mathbf{x},t) = \sum_{i=0}^{p-1} u_i(\mathbf{x},t). \tag{7}$$

The partial sums defined in (7) help to approximate the original solution. Note that the evaluation of (6) requires sufficient smoothness of the initial condition in order to evaluate $R(u)$. It is observed that this method is effective for most of the initial value problems (Jafari & Gejji, 2005) when the space domain is unbounded. However, the effectiveness of this method is open for partial differential equations on a bounded domain. We will provide a sufficient condition in favor of this in the present work in the upcoming section. In this paper, ADM is expected to have time singular functions starting from initial approximations. We expect that the ADM-PINN combination will provide a better approximation than ADM. Therefore, we first discuss the combination of the ADM-PINN.

3.2. ADM-PINN combination procedures

Eq. (1) admits two types of classifications, depending on the value of α. The case $\alpha = 1$ is easy to handle by automatic differentiation in the machine learning algorithm. Our interest lies in the range $\alpha \in (0,1)$, for which standard PINN approaches based on automatic differentiation are not directly applicable. The traditional neural network algorithm serves the classical chain rule in automatic differentiation, which is used to find the derivatives of the defined model, and gradient descent is used to optimize the model prediction. Therefore, automatic differentiation has several challenges in finding the fractional derivatives. So, first, we discretize the time-fractional derivative using the renowned $L1$ scheme over the uniform mesh.

We choose $t_n = nT/N$, and $\Delta t_n = t_n - t_{n-1}$ for $n = 1, 2, \cdots, N$, $\Delta t = \max \Delta t_n$. Let $\mathcal{T}^N := \{t_n : 0 \le n \le N\}$ be the time discretization and $\mathscr{M}^N = \{(\mathbf{x}, t_n) : \mathbf{x} \in \overline{\Omega}_{\mathbf{x}},\ t_n \in \mathcal{T}\}$. Let us denote $D_N^\alpha u$; as the L1 approximation of $D_t^\alpha u$ at $(\mathbf{x}, t_n)$; defined as follows:

$$D_N^\alpha u(\mathbf{x},t_n) := \mathcal{D}_{n,1} u(\mathbf{x},t_n) - \mathcal{D}_{n,n} u(\mathbf{x},t_0) + \sum_{k=1}^{n-1} (\mathcal{D}_{n,k+1} - \mathcal{D}_{n,k}) u(\mathbf{x},t_{n-k}), \tag{8}$$

where

$$\mathcal{D}_{n,k} = \frac{(t_n - t_{n-k})^{1-\alpha} - (t_n - t_{n-k+1})^{1-\alpha}}{\Gamma(2-\alpha)\cdot \Delta t_{n-k+1}}, \quad 0 \le \mathcal{D}_{n,k+1} \le \mathcal{D}_{n,k}. \tag{9}$$

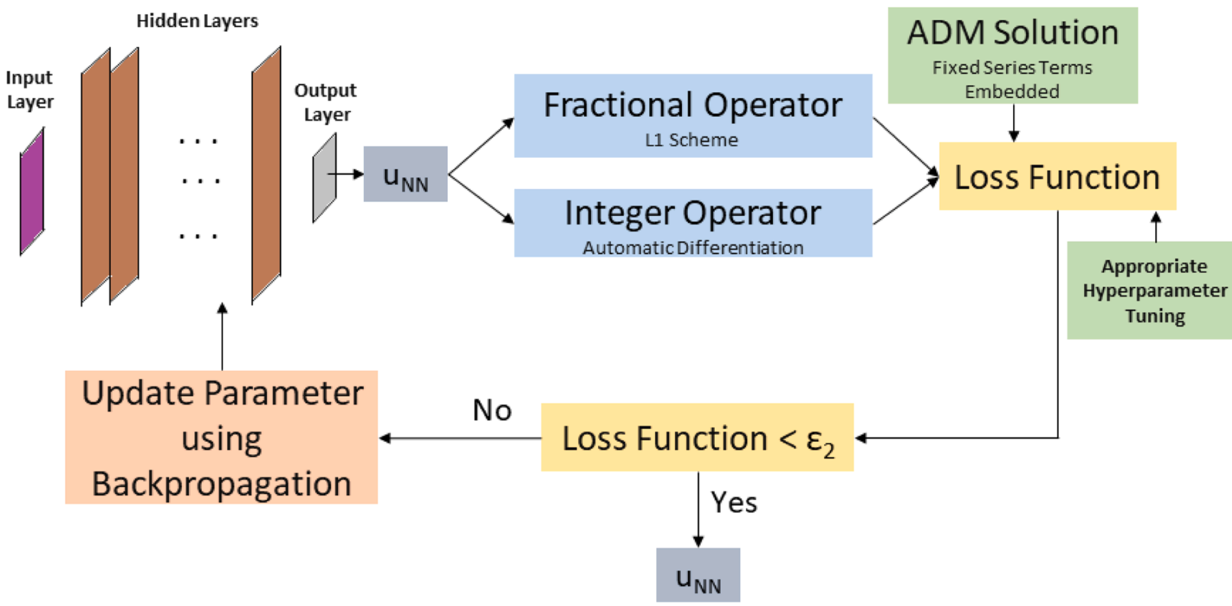


Fig. 1. An image of a proposed feed-forward fractional neural network.

Based on the above discretization (8), we can write the semi-discrete version of (1) over $\iiint^N$, as follows:

$$\mathcal{D}_{n,1}\tilde{u}(\mathbf{x},t_n) = a\,\Delta\tilde{u}(\mathbf{x},t_n) - b(\mathbf{x},t_n)\,\tilde{u}(\mathbf{x},t_n) + f(\mathbf{x},t_n) + \mathcal{D}_{n,n}\tilde{u}(\mathbf{x},t_0)$$
$$- \sum_{k=1}^{n-1}(\mathcal{D}_{n,k+1} - \mathcal{D}_{n,k})\tilde{u}(\mathbf{x},t_{n-k}), \tilde{u}(\mathbf{x},0) = s(\mathbf{x}), \quad \tilde{u}(\mathbf{x},t_n) = 0, \quad n = 1,2,\cdots,N, \tag{10}$$

where $\tilde{u}$ is the semi discretize solution. The $L1$ semi-discretization is useful to find the fractional derivative through the neural network algorithm. Therefore, we include PINN with the semi-analytic series approximation to obtain a better prediction by training the neural nets highlighted in Fig. 1. To find an approximate solution u_{NN} for (10), first, we employ a neural network function $\hat{u}(\mathbf{x},t,\theta)$, which is estimated with the help of affine transformations, activation function, and the neural network structure as follows

$$\hat{u}(\mathbf{x},t;\theta) = A_{k+1}\circ\sigma\circ A_k\circ\cdots A_2\circ\sigma\circ A_1(X^0), \quad k = 1,2,\cdots,K, \tag{11}$$

where A_k's are the affine transformation of neural nets. $A_k : \mathbb{R}^{n_k} \to \mathbb{R}^{n_{k+1}}$, where n_k denotes the neurons in the k^{th} layer of the neural structure, σ is the activation function, $\theta = \{w, b\}$ denotes the parameter inside the neural nets. These affine transformations take X^0 as input. This initial guess helps the optimizer to obtain an optimal parameter θ^* of the network for the optimal prediction through the neural network. The approximated solution of neural network is expressed as follows:

$$u_{NN}(\mathbf{x},t;\theta) = \hat{u}(\mathbf{x},t;\theta). \tag{12}$$

Now, we define the residual error functions corresponding to the neural network solution u_{NN} over the domain $\iiint^N$, as follows

$$R_{pde}(\mathbf{x},t_n) = D_N^{\alpha}u_{NN}(\mathbf{x},t_n) - a\Delta u_{NN}(\mathbf{x},t_n) + b(\mathbf{x},t_n)u_{NN}(\mathbf{x},t_n) - f(\mathbf{x},t_n),$$
$$R_{int}(\mathbf{x},0) = u_{NN}(\mathbf{x},0) - s(\mathbf{x}), \quad R_{bd}(\mathbf{x},t_n) = u_{NN}(\mathbf{x},t_n).$$

We use these residual functions for defining the following loss function to train the neural network for the semi-discrete problem (10)

$$\mathcal{L}(\mathcal{J}) := \lambda_1\mathcal{L}_{pde} + \lambda_2\mathcal{L}_{int} + \lambda_3\mathcal{L}_{bd} + \lambda_4\mathcal{L}_{ADM}, \tag{13}$$

where $\mathcal{L}_{pde}$, $\mathcal{L}_{int}$, and $\mathcal{L}_{bd}$, $\mathcal{L}_{ADM}$ are defined by the following ways

$$\mathcal{L}_{pde} = \sum_{i=1}^{M_{pde}} w_i^{pde}|R_{pde}(\mathbf{x}_{pde}^i, t_{pde}^n)|^2, \mathcal{L}_{int} = \sum_{i=1}^{M_{int}} w_i^{int}|R_{int}(\mathbf{x}_{int}^i, 0)|^2,$$
$$\mathcal{L}_{bd} = \sum_{i=1}^{M_{bd}} w_i^{bd}|R_{bd}(\mathbf{x}_{bd}^i, t_{bd}^n)|^2,$$
$$\mathcal{L}_{ADM} = \sum_{i=1}^{M_{data}} w_i^{data}|u_{NN}(\mathbf{x}_{data}^i, t_{data}^n) - u_{ADM}(\mathbf{x}_{data}^i, t_{data}^n)|^2,$$
$$n = 1,2,\cdots,N.$$

Here M_{pde}, M_{int}, and M_{bd} are the number of partition points at a fixed time level over the interior of the domain, the number of space points at the initial time level, and the number of boundary points of the domain at a fixed time level, respectively. Here, M_{data} denotes the total number of labeled points generated through the ADM Algorithm 1, and used in the $\mathcal{L}_{ADM}$ loss function for optimal choices of parameters. The tuning parameters $\lambda_1,\ldots,\lambda_4$ are user-chosen for better approximation. Mishra and Molinaro (2022) used the following quadrature rule to obtain the theoretical bounds of $R_{pde}(\mathbf{x},t_n)$, $R_{int}(\mathbf{x},0)$, and $R_{bd}(\mathbf{x},t_n)$ for $n = 1,2,\cdots,N$. For a function $O \in C^k(\partial\Omega_{\mathbf{x}})$ at a fixed time level, the quadrature rule corresponding to quadrature weights w_i^{bd} at points $(\mathbf{x}_i,t_n)$, with $1 \leqslant n \leqslant N$ and $1 \leqslant i \leqslant M_{\mathrm{bd}}$, satisfies

$$\left|\int_{\partial\Omega_{\mathbf{x}}} O(\mathbf{x},t_n)\mathrm{d}s(\mathbf{x}) - \sum_{i=1}^{M_{\mathrm{bd}}} w_n^{\mathrm{bd}}O(x_i,t_n)\right|$$
$$\leqslant C_{\mathrm{quad}}^{\mathrm{bd}}\left(\|O\|_{C^k}\right)M_{bd}^{-\beta}, \ \beta \geq 1, \quad n = 1,2,\cdots,N.$$

Similarly, for a function $O \in C^l(\Omega_{\mathbf{x}})$, the quadrature rule corresponding to quadrature weights w_i^{pde} at points $(\mathbf{x}_i,t_n)$, with $1 \leqslant n \leqslant N$ and $1 \leqslant i \leqslant M_{\mathrm{pde}}$, satisfies

$$\left|\int_{\Omega_{\mathbf{x}}} O(\mathbf{x},t_n)\mathrm{d}\mathbf{x} - \sum_{i=1}^{M_{\mathrm{pde}}} w_n^{\mathrm{pde}}O(\mathbf{x}_i,t_n)\right|$$
$$\leqslant C_{\mathrm{quad}}^{\mathrm{pde}}\left(\|O\|_{C^{\ell}}\right)M_{\mathrm{pde}}^{-\beta}, \ \beta \geq 1, \quad n = 1,2,\cdots,N.$$

Finally, for a function $O \in C^k(\Omega_{\mathbf{x}})$, the quadrature rule corresponding to quadrature weights w_i^{int} at points $\mathbf{x}_i$, with $1 \leqslant i \leqslant M_{\mathrm{int}}$, satisfies

$$\left|\int_{\Omega_{\mathbf{x}}} O(\mathbf{x})\mathrm{d}\mathbf{x} - \sum_{i=1}^{M_{\mathrm{int}}} w_n^{\mathrm{int}}O(\mathbf{x}_i)\right| \leqslant C_{\mathrm{quad}}^{\mathrm{int}}\left(\|O\|_{C^k}\right)M_{\mathrm{int}}^{-\beta}, \ \beta \geq 1.$$

Here, $C^{\text{bd}}_{\text{quad}}$, $C^{\text{pde}}_{\text{quad}}$, $C^{\text{int}}_{\text{quad}}$ are the constants defined by the quadrature error mentioned in the above inequalities, $(\mathbf{x}_i, t_n)$ are the partition points of the corresponding domain at $t = t_n$, chosen uniformly. At any fixed time level, we have $i = 1, 2, \cdots, M_{bd}$ for the boundary of the domain and $i = 1, 2, \cdots, M_{int}$ for the initial state of the domain and $i = 1, 2, \cdots, M_{pde}$ for the interior points of the space domain. In principle, different-order quadrature rules can be used in the above inequalities, for example, see Mishra and Molinaro (2022). Eq. 10 can be written in the following form over the domain $\mathcal{J}\mathcal{J}\mathcal{J}^N$ to find u_{NN} through the neural network as follows:

$$\begin{aligned} D^{\alpha}_N u_{NN}(\mathbf{x}, t_n) &= a\, \Delta u_{NN}(\mathbf{x}, t_n) - b(\mathbf{x}, t_n)\, u_{NN}(\mathbf{x}, t_n) + f(\mathbf{x}, t_n) \\ &\quad + R_{pde}(\mathbf{x}, t_n), \quad (\mathbf{x}, t) \in \Omega_{\mathbf{x}} \times (0, T], \\ u_{NN}(\mathbf{x}, 0) &= s(\mathbf{x}) + R_{int}(\mathbf{x}, 0), \quad \mathbf{x} \in \overline{\Omega}_{\mathbf{x}}, \\ u_{NN}(\mathbf{x}, t_n) &= R_{bd}(\mathbf{x}, t_n), \quad (\mathbf{x}, t) \in \partial\Omega_{\mathbf{x}} \times [0, T], \end{aligned} \tag{14}$$

where $D^{\alpha}_N u_{NN}(\mathbf{x}, t_n)$ is defined as follows:

$$\begin{aligned} D^{\alpha}_N u_{NN}(\mathbf{x}, t_n) &= \mathcal{D}_{n,1} u_{NN}(\mathbf{x}, t_n) - \mathcal{D}_{n,n} u_{NN}(\mathbf{x}, t_0) \\ &\quad + \sum_{k=1}^{n-1} (\mathcal{D}_{n,k+1} - \mathcal{D}_{n,k}) u_{NN}(\mathbf{x}, t_{n-k}). \end{aligned} \tag{15}$$

4. Analysis of decomposition method

Here, we provide the convergence analysis of the ADM-based solution and the sufficient conditions under which the Adomian method will work for time fractional partial differential equations on a bounded domain.

Note that, one can expect a relatively smaller error through the ADM-PINN combination if one chooses a partial sum of ADM solution inside the loss function of the ADM-PINN algorithm.

4.1. Convergence analysis

Now we discuss the convergence of the ADM defined in Section 3.1. Let us consider the Hilbert space $\mathcal{H} = L^2(0, T; L^2(\Omega_{\mathbf{x}}))$, i.e., for $v \in \mathcal{H}$ we have $\int_0^T \int_{\Omega_{\mathbf{x}}} v^2(\mathbf{x}, r) d\mathbf{x} dr < \infty$. For $v, w \in \mathcal{H}$, the inner product and the associated norm are defined by

$$\begin{aligned} < v,\ w >: &= \int_0^T \int_{\Omega_{\mathbf{x}}} v(\mathbf{x}, r) w(\mathbf{x}, r) d\mathbf{x} dr \quad \text{and} \quad \|v\|^2_{L^2(0,T;L^2(\Omega_{\mathbf{x}}))} = \|v\|^2 \\ &: = \int_0^T \int_{\Omega_{\mathbf{x}}} v^2(\mathbf{x}, r) d\mathbf{x} dr. \end{aligned}$$

Now, let us assume $L(u) = D^{\alpha}_t u$ and $R(u) = -a\Delta u + b(\mathbf{x}, t)u$ in (4) and consider an operator $F : \mathcal{H} \to \mathcal{H}$ defined by $F(u) = -R(u)$.

Lemma 1. *The operator F is hemicontinuous on $\mathcal{H}$.*

Proof: Let $\{s_n\}_{n\in\mathbb{N}}$ be a sequence of real numbers such that $s_n \to 0$ as $n \to \infty$. For $v \in L^2(0, T; H^2(\Omega_{\mathbf{x}}))$, we have $F(u + s_n v) \in \mathcal{H}$ from Theorem 1. Now, to prove F is hemicontinuous, we have to show that $F(u + s_n v)$ converges weakly to Fu, i.e., for all $w \in \mathcal{H}$,

$$\lim_{n\to\infty} < F(u + s_n v), w > \to < Fu, w > .$$

Again from Theorem (1), $u,\ v \in L^2(0, T; H^2(\Omega_{\mathbf{x}}))$ imply $\Delta u,\ \Delta v \in L^2(0, T; L^2(\Omega_{\mathbf{x}}))$ and also $u,\ v \in L^2(0, T; L^2(\Omega_{\mathbf{x}}))$. Note that $F(u + s_n v) = a\Delta(u + s_n v) - b(\mathbf{x}, t)(u + s_n v)$. Hence, for all $w \in \mathcal{H}$, we have

$$\begin{aligned} &< F(u + s_n v) - F(u), w >=< s_n(a\Delta v - b(\mathbf{x}, t)v), w > \\ &= s_n \int_0^T \int_{\Omega_{\mathbf{x}}} (a\Delta v(\mathbf{x}, r) - b(\mathbf{x}, r)v(\mathbf{x}, r))\ w(\mathbf{x}, r) dr d\mathbf{x}. \end{aligned} \tag{16}$$

Since $v, w, \Delta v \in \mathcal{H}$, we have $\int_0^T \int_{\Omega_{\mathbf{x}}} (a\Delta v(\mathbf{x}, r) - b(\mathbf{x}, r)v(\mathbf{x}, r))w(\mathbf{x}, r) dr d\mathbf{x} < \infty$ i.e., the integral term is bounded. Hence, by taking the limit $n \to \infty$ at both sides of (16), we get

$$\lim_{n\to\infty} < F(u + s_n v) - F(u), w > \to\ 0,$$

which implies $< F(u + s_n v), w >$ converges to $< F(u), w >$ weakly as $n \to \infty$. Now we discuss a lemma, which is used to prove our convergence theorem.

Lemma 2. *Let $u \in L^2(0, T; H^1_0(\Omega_{\mathbf{x}}))$ then there exists a real number $\beta > 0$, $\langle \Delta u, u \rangle \geq \beta \|u\|^2$.*

Proof. By using the definition of inner product we have

$$\langle \Delta u, u \rangle = \int_0^T \int_{\Omega_{\mathbf{x}}} \Delta u\ u\ d\mathbf{x} dt = \int_0^T \int_{\Omega_{\mathbf{x}}} |\nabla u|^2\ d\mathbf{x}\ dt = \|\nabla u(t)\|^2.$$

Now we use the Poincare inequality for a function $u \in L^2(0, T; H^1_0(\Omega_{\mathbf{x}}))$, which assures the existence of a real number $\beta > 0$ such that

$$\|\nabla u\|^2 \geq \beta\ \|u\|^2.$$

From the Poincare inequality, the above result follows. □

Lemma 3. *If T is a linear operator on a Hilbert Space $\mathcal{H}$ which is symmetric in the sense of the inner product of the Hilbert Space i.e.,*

$$\langle Tu, v \rangle = \langle u, Tv \rangle,$$

for all $u, v \in \mathcal{H}$, then T is continuous.

See more details of Lemma 3 in Rudin (1991).

Now we show that the present choice of F satisfies the following hypothesis (Mavoungou & Cherruault, 1992), which is required for the convergence of the Adomian method:

• $(h1) < F(u) - F(v), u - v > \ \geq C_1 \|u - v\|^2$; for some $C_1 > 0$, and for all $u, v \in \mathcal{H}$.

• $(h2)$ For all $M > 0$, for $u, v \in \mathcal{H}$ such that $\|u\| \ \leq M$, $\|v\| \leq M$, there exists a constant $C_2(M) > 0$ such that

$$< F(u) - F(v), w > \leq C_2(M)\|u - v\|\ \ \|w\|, \quad \text{for all} \quad w \in \mathcal{H}.$$

These conditions are sufficient (Cherruault, 1989) for the convergence of the Adomian method.

Eq. (4) can be written as $u - \mathcal{S}u = g$ where $\mathcal{S} = L^{-1}(R)$ and $g = L^{-1} f + \theta$ where $L\theta = 0$. Assuming the above hypothesis $h1$ and $h2$, we have the following theorem, which states the convergence of the Adomian approximation.

Theorem 2. *The following statements hold true.*
• *For every $g \in \mathcal{H}'$ where $\mathcal{H}'$ is dual of $\mathcal{H}$, there exists $u \in \mathcal{H}$ such that $u - \mathcal{S}u = g$.*
• *Let u_n be the sequence defined by the relation $u_{n+1} = u_n - \varsigma\mathcal{S}(u_0 + u_n)$, $\varsigma > 0$, is strongly convergent in $\mathcal{H}$ and the limit u is the solution of $u = \mathcal{S}(u_0 + u)$ for a well chosen $\varsigma > 0$.*

Proof. The proof of this theorem is given in Cherruault (1989).

Hence, it is enough to show that the above hypothesis $(h1) - (h2)$ satisfied for the present model, which follows from the following theorem. □

Theorem 3. *If the above hypothesis $(h1)$ and $(h2)$ are satisfied and the operator F is hemicontinuous then the decomposition method (4) applied to (1) will converge to the exact solution.*

Proof. By Lemma 1, the operator $F(u) = a\Delta u - b(\mathbf{x}, t)u$ is hemicontinuous. Now, we will verify hypothesis $(h1)$ and $(h2)$.
For hypothesis $(h1)$: Let $u, v \in \mathcal{H}$, we get,

$$\langle F(u) - F(v), (u - v) \rangle = \langle a\ \Delta(u - v), (u - v) \rangle - \ \langle b(\mathbf{x}, t)(u - v), (u - v) \rangle.$$

By Lemma 2 for $(u - v) \in L^2(0, T; H^1_0(\Omega_{\mathbf{x}}))$, there exists a real number $\beta > 0$, we have

$$\langle a\ \Delta(u - v), (u - v) \rangle\ \geq a\beta \|u - v\|^2.$$

Again, by Schwartz inequality, we get for $u, v \in \mathcal{H}$,

$$\left\langle b(\mathbf{x},t)(u-v),(u-v)\right\rangle \leq \|b\|_\infty \|u-v\|^2,$$

which implies

$$-\left\langle b(\mathbf{x},t)(u-v),(u-v)\right\rangle \geq -\|b\|_\infty \|u-v\|^2.$$

Therefore, we have

$$\left\langle F(u)-F(v),(u-v)\right\rangle \geq (a\beta - \|b\|_\infty)\|u-v\|^2.$$

Assuming $C_1 = a\beta - \|b\|_\infty$ with $a\beta > \|b\|_\infty$, we verify $(h1)$.
For hypothesis $(h2)$: For all $M > 0$, and $u, v \in \mathcal{H}$ we consider $\|u\| \leq M, \|v\| \leq M$. Again, for $w \in \mathcal{H}$ using Schwartz inequality and Lemma 3, we have

$$\begin{aligned}\left\langle F(u)-F(v),w\right\rangle &= a\left\langle \Delta(u-v),w\right\rangle + \left\langle -b(\mathbf{x},t)(u-v),w\right\rangle \\ &\leq \bar{a}\|u-v\| \;\; \|w\| + \|b\|_\infty \;\|u-v\| \;\; \|w\|.\end{aligned}$$

Therefore, by setting $C_2(M) = \bar{a} + \|b\|_\infty > 0$, we get

$$\left\langle F(u)-F(v),w\right\rangle \leq C_2(M)\|u-v\| \;\; \|w\|.$$

Hence, the hypothesis $(h2)$ is verified. Therefore, we can conclude from [Theorem 4.1, Cherruault (1989)], that the approximate solution based on (6) will converge to the exact solution of (1). □

4.2. Sufficient conditions on the data

In this subsection, we derive a sufficient condition under which the Adomian method for PDEs having initial and boundary conditions works well. To explain this, let us start with the following boundary value problem in the ODE case:

$$\begin{aligned}&Lu \equiv u''(x) = b(x)\,u(x) + f(x), \;\; x \in (\gamma, \delta), \\ &u(\gamma) = A, \;\; u(\delta) = B,\end{aligned} \tag{17}$$

where $b(x) > 0$ is sufficiently smooth function in $[\gamma, \delta]$. By following the procedure in (6) based on ADM, we get

$$u(x) = u(\gamma) + xu'(\gamma) + L^{-1}(b(x)\,u(x) + f(x)). \tag{18}$$

Here, $u'(\gamma)$ is not known. Substituting δ in (18), we get

$$u(\delta) = u(\gamma) + \delta u'(\gamma) + L^{-1}(b(x)\,u(x) + f(x))|_{x=\delta}. \tag{19}$$

Using (19), we obtain $u'(\gamma)$ which can be substituted in (18) to get an approximate solution of (17) when the boundary conditions are specified. The above procedure based on initial data can be used for PDEs with unbounded space domain (Abdelrazec & Pelinovsky, 2011; Dehghan et al., 2007). However, it is not usual for partial differential equations with bounded domains since the independent variables are more.

Now, we provide a sufficient condition on the given data under which the Adomian method will also work for PDEs on bounded domains.

4.2.1. Compatibility of the data

Here we derive the sufficient conditions for the compatibility of the initial and boundary data in (1) on $\{\overline{\Omega}_{\mathbf{x}} \times \{t=0\}\} \cup \{\partial\Omega_{\mathbf{x}} \times [0,T]\}$ for the classical case $\alpha = 1$. This will make the smooth solution at the boundary. For simplicity, we consider the reaction coefficient b as a function of only t. Later, we discuss this for fractional order α.

By means of zeroth-order compatibility at $t = 0$, we want the initial data $s(\mathbf{x})$ must match the homogeneous boundary data, i.e., $s(\mathbf{x}) = 0$ on $\partial\Omega_{\mathbf{x}}$. Similarly, the first order compatibility condition requires the following relation between $f(\mathbf{x},t)$ and $s(\mathbf{x})$:

$$(a\Delta - b(t))s(\mathbf{x}) + \; f(\mathbf{x},0) = 0 \text{ on } \partial\Omega_{\mathbf{x}}. \tag{20}$$

The above relation can be simpler if $s(\mathbf{x}) = 0$ is provided on the boundary.

To obtain the second order compatibility condition, we differentiate (1) with respect to time and obtain the following relation at $t = 0$ by using initial and boundary data; and the above relation

$$\begin{aligned}&a^2\Delta^2 s(\mathbf{x}) - 2\,ab(0)\Delta s(\mathbf{x}) + (a\Delta f(\mathbf{x},t) - b(t)f(\mathbf{x},t) \\ &\qquad + (D_t^1 f(\mathbf{x},t)))|_{t=0} = 0 \text{ on } \partial\Omega_{\mathbf{x}}.\end{aligned} \tag{21}$$

Similarly, third order compatibility condition can be obtained by taking time derivatives of order $\alpha = 1$ two times of (1) with the initial and boundary data at $t = 0$:

$$\begin{aligned}&a^3\Delta^3 s(\mathbf{x}) - 3\,a^2 b(0)\Delta^2 s(\mathbf{x}) + 3\,a(b^2(0) - b_t(0))\Delta s(\mathbf{x}) \\ &\qquad + a^2\;\Delta^2 f(\mathbf{x},0) - 2\,a\Delta b(0) f(\mathbf{x},0) \\ &\qquad + (b^2(t)f(\mathbf{x},t) - 2D_t^1 b(t) f(\mathbf{x},t) - b(t)D_t^1 f(\mathbf{x},t) + aD_t^1\Delta f(\mathbf{x},t) \\ &\qquad + D_t^2 f(\mathbf{x},t))|_{t=0} = 0 \text{ on } \partial\Omega_{\mathbf{x}}.\end{aligned} \tag{22}$$

The above process can be repeated to obtain the higher-order compatibility conditions for sufficiently smooth initial data $s(\mathbf{x})$, and $b(t)$, $f(\mathbf{x},t)$. Note that, it is sufficient to use only the initial condition for solving (1) if the compatibility conditions of infinite order are satisfied by the initial data, as then the boundary conditions stated in (1) will be automatically satisfied.

Now we obtain sufficient conditions for the compatibility of the initial and boundary data in (1) for fractional order α. For simplicity, we consider the reaction coefficient b as a constant.

Zeroth-order compatibility at $t = 0$, is obtained from the initial data $s(\mathbf{x})$ and the homogeneous boundary data, i.e., $s(\mathbf{x}) = 0$ on $\partial\Omega_{\mathbf{x}}$. Similarly, first order compatibility condition requires the following relation between $f(\mathbf{x},t)$ and $s(\mathbf{x})$:

$$(a\Delta - b)s(\mathbf{x}) + \; f(\mathbf{x},0) = 0 \text{ on } \partial\Omega_{\mathbf{x}}. \tag{23}$$

To obtain the second order compatibility condition, we differentiate (1) fractionally of order α in time and obtain the following relation at $t = 0$ by using initial and boundary data

$$(a\Delta - b)^2 s(\mathbf{x}) + (a\;\Delta - b)f(\mathbf{x},t) + (D_t^\alpha f(\mathbf{x},t)))|_{t=0} = 0 \text{ on } \partial\Omega_{\mathbf{x}}. \tag{24}$$

Similarly, the third order compatibility condition can be obtained by two times taking time fractional derivatives of order α of (1) and, using the initial and boundary data at $t = 0$:

$$\begin{aligned}&(a\Delta - b)^3 s(\mathbf{x}) + ((a\Delta - b)^2 f(\mathbf{x},t) + (a\Delta - b)(D_t^\alpha f(\mathbf{x},t)) \\ &\qquad + (D_t^{2\alpha} f(\mathbf{x},t)))|_{t=0} = 0 \text{ on } \partial\Omega_{\mathbf{x}}.\end{aligned} \tag{25}$$

By repeating the above process, we can obtain the nth order compatibility conditions for given sufficiently smooth initial data s, and b, f:

$$\begin{aligned}&(a\Delta - b)^n s(\mathbf{x}) + ((a\Delta - b)^{n-1} f(\mathbf{x},t) + (a\Delta - b)^{n-2}(D_t^\alpha f(\mathbf{x},t)) + \cdots \\ &\qquad + (a\Delta - b)(D_t^{(n-2)\alpha} f(\mathbf{x},t)) \\ &\qquad + (D_t^{(n-1)\alpha} f(\mathbf{x},t)))|_{t=0} = 0 \text{ on } \partial\Omega_{\mathbf{x}}.\end{aligned} \tag{26}$$

Note: If $f(\mathbf{x},t)$ is the form of the $k(t)s(\mathbf{x})$, where k is any smooth function of t, infinite order compatibility conditions hold without solving any large number of equations for integer order as well as fractional order operators. Before going further, we provide the key steps of the ADM algorithm as follows: Note that for our ADM-PINN algorithm, we take

Algorithm 1 ADM Algorithm for solution approximation.

Input: Tolerance ϵ_1.
Output: $\Phi_*(\mathbf{x},t)$.
Step 1. Find u_p with $p = 0$, by (6).
Step 2. Assume $\Phi_0(\mathbf{x},t) = 0$. Calculate the approximate solution $\Phi_p(\mathbf{x},t) = \Phi_{p-1}(\mathbf{x},t) + u_{p-1}(\mathbf{x},t)$.
Step 3. Check $||L(\Phi_p(\mathbf{x},t)) + R(\Phi_p(\mathbf{x},t)) - f(\mathbf{x},t)|| < \epsilon_1$.
Step 4. If not, take $p = p + 1$.
Step 5. Repeat Step 1 with $p = p + 1$.
Step 6. Desired approximate solution will be $\Phi_*(\mathbf{x},t)$ from Step 2.

the approximate solution from the above algorithm with a fixed value of j from Step 2, for generating an appropriate prediction.

5. Estimation of the generalization error

Let us first define the ADM-based PINN algorithm for the approximation of neural network solution when the initial guess is provided by the semi-analytic approximation. We have defined a convergence analysis for the ADM-based series solution in Section 4. The minimum number of terms after which the partial sum-based ADM solution will start to decrease at any partition points is not clear from theory. Hence, we start with any ADM-based partial sum as an initial approximation for the above algorithm before utilizing PINN for improvement and convergent results. This algorithm is specified and expected to generate a better solution as it starts with ADM approx., i.e., a fixed partial sum, which can detect the time singularity.

Algorithm 2 ADM-PINN Algorithm for solution approximation of (1).

Input: Domain (Ω), Learning rate (η), Tolerance (ϵ_2), Iteration ($iter$), Activation function (σ), Depth (K), Loss function (say, $\mathcal{L}(\theta)$ as defined in (13), $\theta = \{w, b\}$), Training points (M_{pde}, M_{bd}, M_{int}).
Output: u^p_{NN} (approximate solution based on neural network)
Semi-discretize (1) through $L1$ scheme on the temporal direction. ▷ See (10)
Find ADM approximation (u_{ADM}) mentioned in Algorithm 1 with $p = P_1$ ▷ will be used in (13).
Initializing parameters (θ) as $\theta = \{w, b\}$
$u^p_{NN}(\mathbf{x}, t, \theta) = [\ ]$, ▷ Defining storage
for *($m = 1$ to $iter$)* **do**
  for $k = 1, \cdots, K-1$ **do**
    $A^k = w^k X^{k-1} + b^k$, ▷ X^0 depends on training points
    $X^k = \sigma(A^k)$, ▷ σ is taken as Tanh
  end
  $A^K = w^K X^{K-1} + b^K$,
  $X^K = A^K$,
  $\hat{u}(\mathbf{x}, t, \theta) = X^K$,
  $u^p_{NN}(\mathbf{x}, t, \theta) = \hat{u}(\mathbf{x}, t, \theta)$
  Evaluate loss function $\mathcal{L}(\theta)$ as defined in (13)
  if $\mathcal{L}(\theta) < \epsilon_2$ **then**
    break ▷ Early stopping to prevent overfitting
  end
  Compute $\nabla_\theta(\mathcal{L}(\theta))$ through back propagation
  Update θ
end
$u^p_{NN}(\mathbf{x}, t, \theta)$ ▷ Final approximation after all iteration

We are interested in estimating the bound of the generalization error for a fractional version of PINN. According to (1) and (14), we can easily write the error $\phi = u - u^p_{NN}$ for the present model as follows

$$\phi = u - u^p_{NN} = \underbrace{(u - \tilde{u})}_{\phi_1} + \underbrace{(\tilde{u} - u^p_{NN})}_{\phi_2^p}.$$

The error bounds of ϕ_1, ϕ_2^p will be derived separately.

We have the following lemma for estimating the error for a semi-discrete solution, which is useful to derive a generalization error.

Lemma 4. *Stynes et al. (2017) $u \in C^{4,2}(\bar{\Omega}_{\mathbf{x}} \times [0, T])$ satisfy the given smoothness condition. Let $\tilde{u}(\mathbf{x}, t_n)$ be a solution of semi-discrete problem (10). Then, for every $(\mathbf{x}, t_n) \in \iiint^N$, we have the following error bound for ϕ_1 for uniform mesh:*

$$\|\phi_1(\mathbf{x}, t_n)\|_{L^2(\bar{\Omega}_{\mathbf{x}})} \le C n^{-(2-\alpha)}.$$

Now, we need to find the error bound corresponding to ϕ_2^p. To find this, we need the following discrete form of Gronwall lemma.

Lemma 5. *Sloan and Thomée (1986) Let Θ_n be a sequence of non-negative real numbers, which satisfies*

$$\Theta_n \le \gamma_n + \sum_{k=0}^{n-1} \chi_k \Theta_k, \quad n \ge 0,$$

where, γ_n is a sequence of non-decreasing and non-negative numbers. Assume $\chi_k \ge 0$. Then, the following relation holds true, for $n \ge 0$,

$$\Theta_n \le \gamma_n \left(\exp \sum_{k=0}^{n-1} \chi_k \right).$$

We need an error equation to find the error bound of a semi-discrete neural network solution. To define the error equation in terms of ϕ_2^p, we use (10) and (14), and define it as follows

$$\begin{aligned} D^\alpha_N \phi_2^p(\mathbf{x}, t_n) &= a\, \Delta \phi_2^p(\mathbf{x}, t_n) - b(\mathbf{x}, t_n)\, \phi_2^p(\mathbf{x}, t_n) \\ &\quad + R^p_{pde}(\mathbf{x}, t_n), \quad (\mathbf{x}, t) \in \Omega_{\mathbf{x}} \times (0, T], \\ \phi_2^p(\mathbf{x}, 0) &= R^p_{int}(\mathbf{x}, 0), \quad \text{on } \mathbf{x} \in \overline{\Omega}_{\mathbf{x}}, \\ \phi_2^p(\mathbf{x}, t_n) &= R^p_{bd}(\mathbf{x}, t_n), \quad \text{on } (\mathbf{x}, t) \in \partial\Omega_{\mathbf{x}} \times [0, T]. \end{aligned} \tag{27}$$

Remark 1. In the ADM-PINN framework, the series solution u_{ADM} obtained from the ADM is incorporated into the loss function. To enhance the accuracy of the PINN solution, we employ a weighted-loss hyperparameter strategy. Initially, a relatively small weight is assigned to the ADM-based loss term. As the number of series terms increased in the u_{ADM} then the weight also increased. This progressive weighting mechanism enhances the overall accuracy and stability of the PINN solution. In the upcoming theorem, we will see the impact of the series terms p on the generalization error.

Remark 2. In the theoretical analysis of the decomposition method, we proved that the residual error of the partial differential equation decreases as the number of series terms increases. Since the ADM data is used in the loss function to enhance the accuracy of the neural network solution. Therefore, the residual error of the ADM-PINN is also decreasing as the number of series terms increases.

Theorem 4. *Let $u \in C^{4,2}(\bar{\Omega}_{\mathbf{x}} \times [0, T])$ be the solution of (1). Let u^p_{NN} be a neural network solution of (14) generated through Algorithm 2, corresponding to the loss function (13). Then, the corresponding error can be estimated as follows*

$$\begin{aligned} \epsilon^2(n) = \int_{\Omega_{\mathbf{x}}} |\phi_2^p(\mathbf{x}, t_n)|^2 dx \le \Bigg[\frac{4}{3} \int_{\Omega_{\mathbf{x}}} |R^p_{int}(\mathbf{x}, t_0)|^2 d\mathbf{x} + \frac{16}{3\mathcal{D}^2_{n,1}} \int_{\Omega_{\mathbf{x}}} |R^p_{pde}(\mathbf{x}, t_n)|^2 d\mathbf{x} \\ + \frac{C_1}{\mathcal{D}_{n,1}} \left(\int_{\partial\Omega_{\mathbf{x}}} |R^p_{bd}(\mathbf{x}, t_n)|^2 ds(\mathbf{x}) \right)^{1/2} \Bigg] \times \exp\left(\frac{4}{3\mathcal{D}_{n,1}} (\mathcal{D}_{n,1} - \mathcal{D}_{n,n}) \right), \end{aligned} \tag{28}$$

and C_1 constant is defined as:

$$C_1 = |\partial\Omega_{\mathbf{x}}|^{\frac{1}{2}} \left[\|\tilde{u}\| + \|u_{NN}\| \right].$$

Proof. To compute the bound of the generalization error, we multiply $\phi_2^p(\mathbf{x}, t_n)$ on both sides of (27) and integrate over the spatial domain $\Omega_{\mathbf{x}}$, as follows

$$\begin{aligned} \int_{\Omega_{\mathbf{x}}} \phi_2^p(\mathbf{x}, t_n) D^\alpha_N \phi_2^p(\mathbf{x}, t_n) d\mathbf{x} &= \int_{\Omega_{\mathbf{x}}} \phi_2^p(\mathbf{x}, t_n) a\, \Delta \phi_2^p(\mathbf{x}, t_n) d\mathbf{x} \\ &- \int_{\Omega_{\mathbf{x}}} \phi_2^p(\mathbf{x}, t_n) b(\mathbf{x}, t_n)\, \phi_2^p(\mathbf{x}, t_n) d\mathbf{x} + \int_{\Omega_{\mathbf{x}}} \phi_2^p(\mathbf{x}, t_n) R^p_{pde}(\mathbf{x}, t_n) dx. \end{aligned}$$

Using (15), we have

$$\begin{aligned} \mathcal{D}_{n,1} \int_{\Omega_{\mathbf{x}}} \phi_2^p(\mathbf{x}, t_n) \phi_2^p(\mathbf{x}, t_n) d\mathbf{x} &= a \int_{\Omega_{\mathbf{x}}} \phi_2^p(\mathbf{x}, t_n) \Delta \phi_2^p(\mathbf{x}, t_n) d\mathbf{x} \\ &- \int_{\Omega_{\mathbf{x}}} b(\mathbf{x}, t_n) |\phi_2^p(\mathbf{x}, t_n)|^2 d\mathbf{x} + \mathcal{D}_{n,n} \int_{\Omega_{\mathbf{x}}} \phi_2^p(\mathbf{x}, t_n) \phi_2^p(\mathbf{x}, t_0) d\mathbf{x} \\ &+ \int_{\Omega_{\mathbf{x}}} \phi_2^p(\mathbf{x}, t_n) R^p_{pde}(\mathbf{x}, t_n) dx - \int_{\Omega_{\mathbf{x}}} \phi_2^p(\mathbf{x}, t_n) \sum_{k=1}^{n-1} (\mathcal{D}_{n,k+1} - \mathcal{D}_{n,k}) \phi_2^p(\mathbf{x}, t_{n-k}) d\mathbf{x}. \end{aligned}$$

Since $\int_{\Omega_{\mathbf{x}}} b(\mathbf{x},t_n)|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x} \geq 0$, we have

$$\mathcal{D}_{n,1}\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)\phi_2^p(\mathbf{x},t_n)d\mathbf{x} \leq a\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)\Delta\phi_2^p(\mathbf{x},t_n)d\mathbf{x}$$
$$+\mathcal{D}_{n,n}\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)\phi_2^p(\mathbf{x},t_0)d\mathbf{x}+\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)R_{pde}^p(\mathbf{x},t_n)d\mathbf{x}$$
$$-\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)\sum_{k=1}^{n-1}(\mathcal{D}_{n,k+1}-\mathcal{D}_{n,k})\phi_2^p(\mathbf{x},t_{n-k})d\mathbf{x}.$$

Using integration by parts on the above equation, where $\mathbf{n}$ is the unit outward normal of $\Omega_{\mathbf{x}}$, we have

$$\mathcal{D}_{n,1}\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)\phi_2^p(\mathbf{x},t_n)d\mathbf{x} \leq -a\int_{\Omega_{\mathbf{x}}}|\nabla\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x}$$
$$+\int_{\partial\Omega_{\mathbf{x}}}R_{bd}^p(\mathbf{x},t_n)(\nabla\phi_2^p\cdot\mathbf{n})ds(\mathbf{x})$$
$$+\mathcal{D}_{n,n}\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)\phi_2^p(\mathbf{x},t_0)d\mathbf{x}+\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)R_{pde}^p(\mathbf{x},t_n)d\mathbf{x}$$
$$-\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)\sum_{k=1}^{n-1}(\mathcal{D}_{n,k+1}-\mathcal{D}_{n,k})\phi_2^p(x,t_{n-k})d\mathbf{x}.$$

Since $\int_{\Omega_{\mathbf{x}}}|\nabla\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x} \geq 0$, we have

$$\mathcal{D}_{n,1}\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)\phi_2^p(\mathbf{x},t_n)d\mathbf{x} \leq \mathcal{D}_{n,n}\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)\phi_2^p(\mathbf{x},t_0)d\mathbf{x}$$
$$+\int_{\Omega_{\mathbf{x}}}\frac{\mathcal{D}_{n,1}^{1/2}}{2}\phi_2^p(\mathbf{x},t_n)\frac{2}{\mathcal{D}_{n,1}^{1/2}}R_{pde}^p(\mathbf{x},t_n)dx$$
$$+\underbrace{|\partial\Omega_{\mathbf{x}}|^{\frac{1}{2}}\left[\|\tilde{u}\|+\|u_{NN}^p\|\right]}_{C_1}\left(\int_{\partial\Omega_{\mathbf{x}}}(R_{bd}^p)^2(\mathbf{x},t_n)ds(\mathbf{x})\right)^{1/2}$$
$$-\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)\sum_{k=1}^{n-1}(\mathcal{D}_{n,k+1}-\mathcal{D}_{n,k})\phi_2^p(\mathbf{x},t_{n-k})d\mathbf{x}.$$

Using Young's inequality in the above equation, we get

$$\mathcal{D}_{n,1}\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)\phi_2^p(\mathbf{x},t_n)d\mathbf{x} \leq \frac{\mathcal{D}_{n,n}}{2}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x}$$
$$+\frac{\mathcal{D}_{n,n}}{2}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_0)|^2 d\mathbf{x}$$
$$+\frac{\mathcal{D}_{n,1}}{8}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x}+\frac{2}{\mathcal{D}_{n,1}}\int_{\Omega_{\mathbf{x}}}|R_{pde}^p(\mathbf{x},t_n)|^2 d\mathbf{x}$$
$$+C_1\left(\int_{\partial\Omega_{\mathbf{x}}}|R_{bd}^p(\mathbf{x},t_n)|^2 ds(\mathbf{x})\right)^{1/2}$$
$$+\int_{\Omega_{\mathbf{x}}}\phi_2^p(\mathbf{x},t_n)\sum_{k=1}^{n-1}(\mathcal{D}_{n,k}-\mathcal{D}_{n,k+1})\phi_2^p(\mathbf{x},t_{n-k})d\mathbf{x},$$

$$\mathcal{D}_{n,1}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x} \leq \frac{\mathcal{D}_{n,n}}{2}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x}$$
$$+\frac{\mathcal{D}_{n,n}}{2}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_0)|^2 d\mathbf{x}$$
$$+\frac{\mathcal{D}_{n,1}}{8}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x}+\frac{2}{\mathcal{D}_{n,1}}\int_{\Omega_{\mathbf{x}}}|R_{pde}^p(\mathbf{x},t_n)|^2 d\mathbf{x}$$
$$+\int_{\Omega_{\mathbf{x}}}\sum_{k=1}^{n-1}\frac{(\mathcal{D}_{n,k}-\mathcal{D}_{n,k+1})}{2}|\phi_2^p(\mathbf{x},t_{n-k})|^2 d\mathbf{x}$$
$$+\int_{\Omega_{\mathbf{x}}}\frac{(\mathcal{D}_{n,1}-\mathcal{D}_{n,n})}{2}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x}+C_1\left(\int_{\partial\Omega_{\mathbf{x}}}|R_{bd}^p(\mathbf{x},t_n)|^2 ds(\mathbf{x})\right)^{1/2},$$

$$\frac{\mathcal{D}_{n,1}}{2}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x} \leq \frac{\mathcal{D}_{n,n}}{2}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_0)|^2 d\mathbf{x}$$
$$+C_1\left(\int_{\partial\Omega_{\mathbf{x}}}|R_{bd}^p(\mathbf{x},t_n)|^2 ds(\mathbf{x})\right)^{1/2}$$
$$+\frac{\mathcal{D}_{n,1}}{8}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x}+\frac{2}{\mathcal{D}_{n,1}}\int_{\Omega_{\mathbf{x}}}|R_{pde}^p(\mathbf{x},t_n)|^2 d\mathbf{x}$$
$$+\int_{\Omega_{\mathbf{x}}}\sum_{k=1}^{n-1}\frac{(\mathcal{D}_{n,k}-\mathcal{D}_{n,k+1})}{2}|\phi_2^p(\mathbf{x},t_{n-k})|^2 d\mathbf{x}.$$

From (9), the above inequality follows

$$\mathcal{D}_{n,1}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x} \leq \mathcal{D}_{n,n}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_0)|^2 d\mathbf{x}$$
$$+2C_1\left(\int_{\partial\Omega_{\mathbf{x}}}|R_{bd}^p(\mathbf{x},t_n)|^2 ds(\mathbf{x})\right)^{1/2}$$
$$+\frac{\mathcal{D}_{n,1}}{4}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x}+\frac{4}{\mathcal{D}_{n,1}}\int_{\Omega_{\mathbf{x}}}|R_{pde}^p(\mathbf{x},t_n)|^2 d\mathbf{x}$$
$$+\int_{\Omega_{\mathbf{x}}}\sum_{k=1}^{n-1}(\mathcal{D}_{n,k}-\mathcal{D}_{n,k+1})|\phi_2^p(\mathbf{x},t_{n-k})|^2 d\mathbf{x}.$$

Using (9), we get

$$\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x} \leq \frac{4}{3}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_0)|^2 d\mathbf{x}+\frac{16}{3\mathcal{D}_{n,1}^2}\int_{\Omega_{\mathbf{x}}}|R_{pde}^p(\mathbf{x},t_n)|^2 d\mathbf{x}$$
$$+\frac{8C_1}{3\mathcal{D}_{n,1}}\left(\int_{\partial\Omega_{\mathbf{x}}}|R_{bd}^p(\mathbf{x},t_n)|^2 ds(\mathbf{x})\right)^{1/2}$$
$$+\frac{4}{3\mathcal{D}_{n,1}}\sum_{k=1}^{n-1}(\mathcal{D}_{n,k}-\mathcal{D}_{n,k+1})\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_{n-k})|^2 d\mathbf{x},$$

which implies

$$\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x} \leq \gamma_n+\frac{4}{3\mathcal{D}_{n,1}}\sum_{k=1}^{n-1}\chi_k\left(\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_{n-k})|^2 d\mathbf{x}\right).$$

In the above inequality, γ_n includes only the n^{th} time level terms. To show that γ_n is bounded for all time levels, we consider all terms appearing in γ_n, and its bound is defined in Theorem 5.

$$\gamma_n=\frac{4}{3}\int_{\Omega_{\mathbf{x}}}|R_{int}^p(\mathbf{x},t_0)|^2 d\mathbf{x}+\frac{16}{3\mathcal{D}_{n,1}^2}\sum_{i=0}^{n}\int_{\Omega_{\mathbf{x}}}|R_{pde}^p(\mathbf{x},t_i)|^2 d\mathbf{x}$$
$$+\frac{8C_1}{3\mathcal{D}_{n,1}}\left(\sum_{i=0}^{n}\int_{\partial\Omega_{\mathbf{x}}}|R_{bd}^p(\mathbf{x},t_i)|^2 ds(\mathbf{x})\right)^{1/2},\ \chi_k=\mathcal{D}_{n,k}-\mathcal{D}_{n,k+1}.$$

$$\gamma_n=\frac{4}{3}\int_{\Omega_{\mathbf{x}}}|R_{int}(\mathbf{x},t_0)|^2 d\mathbf{x}+\underbrace{\frac{16}{3\mathcal{D}_{n,1}^2}\sum_{i=0}^{n}\int_{\Omega_{\mathbf{x}}}|R_{pde}^p(\mathbf{x},t_i)|^2 d\mathbf{x}}_{I_1}$$
$$+\underbrace{\frac{8C_1}{3\mathcal{D}_{n,1}}\left(\sum_{i=0}^{n}\int_{\partial\Omega_{\mathbf{x}}}|R_{bd}^p(\mathbf{x},t_i)|^2 ds(\mathbf{x})\right)^{1/2}}_{I_2}. \tag{29}$$

From (29), it is obvious that γ_n is a non-negative and non-decreasing sequence. Observe that $\chi_k=\mathcal{D}_{n,k}-\mathcal{D}_{n,k+1}$. From (9), this implies that χ_k is non negative for $k=1,2,\cdots,n-1$.

Since, the γ_n associated with the n^{th} term in the proof, we continue with the n^{th} term in the γ_n. Moreover, the bound of the γ_n for all time levels is provided in Theorem 5. From the discrete form of Gronwall Lemma 5, we get

$$\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x} \leq \left[\frac{4}{3}\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_0)|^2 d\mathbf{x}+\frac{16}{3\mathcal{D}_{n,1}^2}\int_{\Omega_{\mathbf{x}}}|R_{pde}^p(\mathbf{x},t_n)|^2 d\mathbf{x}\right.$$
$$\left.+\frac{8C_1}{3\mathcal{D}_{n,1}}\left(\int_{\partial\Omega_{\mathbf{x}}}|R_{bd}^p(\mathbf{x},t_n)|^2 ds(\mathbf{x})\right)^{1/2}\right]\times\exp\left(\frac{4}{3\mathcal{D}_{n,1}}\sum_{k=1}^{n-1}(\mathcal{D}_{n,k}-\mathcal{D}_{n,k+1})\right).$$

$$\epsilon^2=\int_{\Omega_{\mathbf{x}}}|\phi_2^p(\mathbf{x},t_n)|^2 d\mathbf{x} \leq \left[\frac{4}{3}\int_{\Omega_{\mathbf{x}}}|R_{int}^p(\mathbf{x},t_0)|^2 d\mathbf{x}+\frac{16}{3\mathcal{D}_{n,1}^2}\int_{\Omega_{\mathbf{x}}}|R_{pde}^p(\mathbf{x},t_n)|^2 d\mathbf{x}\right.$$

$$+ \frac{8C_1}{3\mathcal{D}_{n,1}}\left(\int_{\partial\Omega_{\mathbf{x}}} |R^p_{bd}(\mathbf{x}, t_n)|^2 ds(\mathbf{x})\right)^{1/2}\Bigg] \times \exp\left(\frac{4}{3\mathcal{D}_{n,1}}(\mathcal{D}_{n,1} - \mathcal{D}_{n,n})\right).$$

□

From Section 3.1, it is clear that ADM decomposes the exact solution u as $u = \sum_{p=0}^{\infty} u_p$, where u_p is computed recursively using the inverse fractional operator (fractional integral operator) mentioned in (6). Let us consider a class of time-fractional PDEs involving the Caputo derivative (${}^C D_t^{\alpha} m(t),\ 0 < \alpha < 1$), whose solutions exhibit weak singularities in time. Now, we observe the case where the source function of the (1) is smooth, but the exact solution exhibits a weak time singularity. Let $f(\mathbf{x}, t) = \mathfrak{S}(t)\,\mathfrak{F}(\mathbf{x})$, where $\mathfrak{S}(t)$ and $\mathfrak{F}(\mathbf{x})$ are smooth temporal and spatial functions, respectively. If $\mathfrak{F}(\mathbf{x})$ satisfies the initial condition and all even-order derivatives satisfy the boundary, then the ADM-based series solution automatically satisfies the infinite-order compatibility. For the sake of simplicity, let us choose $f = \mathfrak{F}(\mathbf{x})$. In the recurrence relation, the first term of the ADM series solution contains $L^{-1}\mathfrak{F}(\mathbf{x}) = \left(\frac{2t^{\alpha}}{\Gamma(1/2)}\right)\mathfrak{F}(\mathbf{x})$. Hence, the appearance of t^{α} in the series term verifies that ADM can solve the time singularity efficiently. The ADM algorithm uses a fractional integral operator to calculate the series terms. Hence, the integral operator increases the smoothness of the solution and effectively handles its weak singularities even if the source term is non-smooth, whereas the traditional PINN uses the classical chain rule, i.e., not applicable for fractional-order operators. Therefore, ADM component helps to detect the time singularity and converges to the exact solution as series terms increase. In this case, the partial sum in the ADM approximation can be used in the ADM based loss term to detect the time singularity (13). Since the ADM series is convergent, as p increases, the error bound decreases. Hence, the forthcoming Theorem 4 remains valid for fractional problems involving time singularities. We have also seen that ADM uses the series terms to approximate the exact solution and does not depend on any specific partition strategy to compute the approximate solution. Leveraging this advantage, the ADM-PINN algorithm generates the training points through uniform partitions and trains the network efficiently to solve the weak time singularities in the solution.

Remark 3. From (28), it is clear that the generalization error depends on the series terms p of the ADM solution. We have shown that the recurrence relation and the sufficient condition of ADM match the initial condition and boundary conditions; these data help the ADM-PINN algorithm to reduce the error of the boundary and initial conditions based loss function. From Remark 2, the recurrence relation, and the sufficient condition, it is clear that as the number of series terms increases, the generalization error decreases.

Note 1. In the error bound Theorem 4, the nth term of the γ_n appears. Note that this theorem provides error only at nth time level. Hence to calculate the error, accumulated from all time levels, we produce Theorem 5 where the summation over time levels are taken, as it appears inside γ_n. Further, we show that γ_n is a non-negative and non-decreasing sequence, which is useful for the Gronwall lemma, whereas χ_k is non-negative.

Through the defined loss function in (13) and the defined quadrature rule in Mishra and Molinaro (2022), we get the desired inequality (28). Here, we have proved the error is bounded. Since we have used the convergent ADM solution to train the machine learning model, this convergent solution does not harm the bound of the PINN as we increase the number of input data. We get a better solution than ADM if we increase the number of terms in the series expansion of ADM solution to train this model.

The total error can be obtained by adding the semi-discrete based time error and the error obtained in the previous theorem through ADM-PINN. Further, we prove the following lemma and theorem to show the convergence of the estimated error (obtained in above theorem) as a final result in Theorem 5.

Lemma 6. *The term $\chi_k = \mathcal{D}_{n,k} - \mathcal{D}_{n,k+1}$ satisfies $\frac{1}{\mathcal{D}_{n,1}}\sum_{k=1}^{n-1}\chi_k < C_2$ where C_2 is a fixed positive constant.*

Proof. Observe

$$\frac{1}{\mathcal{D}_{n,1}}\sum_{k=1}^{n-1}\chi_k = \frac{1}{\mathcal{D}_{n,1}}\sum_{k=1}^{n-1}(\mathcal{D}_{n,k} - \mathcal{D}_{n,k+1}) = \frac{1}{\mathcal{D}_{n,1}}(\mathcal{D}_{n,1} - \mathcal{D}_{n,n}),$$

which is bounded from (9). □

Remark 4. ADM solution uses the partial sums (7) to approximate the exact solution. Hence, the convergence of the ADM solution is completely based on the series terms; it does not rely on the number of points inside the computational domain. In the upcoming theorem, we will see that the ADM-PINN solution relies on the uniform partition inside the computational domain and decreases as the number of points increases. Here, we are not discussing the convergence based on the number of iterations/epochs in the ADM-PINN algorithm.

Theorem 5. *Let p be a fixed series term. Let us assume the number of uniform partitions in the temporal direction is N and assume the same number of uniform partitions in the spatial directions. This implies M_{pde}, M_{int}, and M_{bd} are also approaching infinity as the uniform step size goes to zero, when $N \to \infty$. Therefore, the error bound $\epsilon(n)$, defined in (28) converges to zero, as $N \to \infty$.*

Proof. We have proved Lemma 6, which is useful for the convergence of (28). Now we show that $\{\gamma_n\}$ satisfies $\gamma_n \to 0$, for all $n = 1, 2, \cdots, N$, as $N \to \infty$. From (29), we have

$$\gamma_n = \frac{4}{3}\int_{\Omega_{\mathbf{x}}} |R^p_{int}(\mathbf{x}, t_0)|^2 d\mathbf{x} + \underbrace{\frac{16}{3\mathcal{D}^2_{n,1}}\sum_{i=0}^{n}\int_{\Omega_{\mathbf{x}}} |R^p_{pde}(\mathbf{x}, t_i)|^2 d\mathbf{x}}_{I_1}$$
$$+ \underbrace{\frac{C_1}{\mathcal{D}_{n,1}}\left(\sum_{i=0}^{n}\int_{\partial\Omega_{\mathbf{x}}} |R^p_{bd}(\mathbf{x}, t_i)|^2 ds(\mathbf{x})\right)^{1/2}}_{I_2}.$$

Now, we prove that I_1 is bounded and goes to zero. Observe

$$\begin{aligned} I_1 &= \frac{16\Delta t_n^{2\alpha}}{3}\sum_{i=0}^{n}\int_{\Omega_{\mathbf{x}}} |R^p_{pde}(\mathbf{x}, t_i)|^2 d\mathbf{x}, \quad n = 1, 2, \cdots, N, \\ &= \frac{16\Delta t_n^{2\alpha}}{3}\left[\int_{\Omega_{\mathbf{x}}} |R^p_{pde}(\mathbf{x}, t_0)|^2 dx + \int_{\Omega_{\mathbf{x}}} |R^p_{pde}(\mathbf{x}, t_1)|^2 d\mathbf{x} + \cdots \right. \\ &\quad \left. + \int_{\Omega_{\mathbf{x}}} |R^p_{pde}(\mathbf{x}, t_n)|^2 d\mathbf{x}\right], \\ &\le \frac{16\Delta t_n^{2\alpha}}{3}\left[C_{quad}\, M_{pde}^{-\beta} + C_{quad}\, M_{pde}^{-\beta} + \cdots + C_{quad}\, M_{pde}^{-\beta}\right], \quad \beta \ge 1, \\ &\le \frac{16\Delta t_n^{2\alpha}}{3} C_{quad}\, M_{pde}^{-\beta}(n+1), \quad \beta \ge 1, \quad n = 1, 2, \cdots, N. \end{aligned}$$

Hence, the above expression is bounded as $N \to \infty$. Also note that $I_1 \to 0$ as $N \to \infty$, as we have chosen the same number of points in both spatial and temporal directions.

Similarly, it can be shown that I_2 is bounded and goes to zero as $N \to \infty$. Hence, $\{\gamma_n\}$ is a non-decreasing and non-negative sequence of numbers such that $\{\gamma_n\}$ goes to zero as $N \to \infty$. Lemma 6 and the above proof are sufficient to provide the convergence of sem discretization based PINN solution. Hence, $\epsilon(n)$ defined in (28) converges to zero, as $N \to \infty$. □

Note 2. We can conclude that the total error estimate of the continuous solution of (1) and the ADM-PINN-based approximate solution can be obtained by adding the error estimate of the time semi-discretization given in Lemma 5 and the error obtained in Theorem 4. Furthermore, the previous theorem demonstrates that this error estimate converges to zero.

Table 1
Hyperparameters employed in the ADM-PINN algorithm during training of the network.

Hyper-parameters	Value
Activation function	$Tanh$
Optimizer	$Adam$
Max epoch	200000
Learning rate	Adaptive
Size of hidden layers	4
Tolerance (ϵ_2)	$1E-5$
No. of neurons in each hidden layer	64

6. Numerical experiments

In this section, we provide several numerical experiments in favor of the ADM-PINN algorithm for time fractional reaction diffusion equations by combining Algorithms 1 and 2. The required hyperparameters of the Algorithm 2 are mentioned in Table 1.

Here, we choose N to be the time partitions, M_x to be the space partitions in the x-direction, and M_y to be the space partitions in the y-direction. The domain discretization over the uniform partitions is defined as follows $\Omega^{m_x,m_y,n} := \{(x_i, y_j, t_n) : m_x = 0, 1, \cdots, M_x, m_y = 0, 1, \cdots, M_y, n = 0, 1, \cdots, N\}$ and generated points (x_i, y_j, t_n) are uniformly distributed in the continuous domain, and assumes $N = M_x = M_y$ throughout the numerical experiments. Further, we will show that the requirement of the epochs decreases as the number of series terms increases in the ADM-PINN algorithm. We perform all experiments using Python 3.12.7, PyTorch 2.4, CUDA 12.1 on an NVIDIA RTX A4000 with 16GB and a host system with 256 GB RAM. To provide the experimental evidence, we begin with the classical problem having an integer-order derivative.

Example 1. Consider the following integer order problem on the space domain $\Omega_{\mathbf{x}} = \Omega_x \times \Omega_y = (0, \pi) \times (0, \pi)$:

$$\begin{cases} D_t^1 u = \frac{1}{5}\ \Delta u - \frac{1}{5}(t^2 + t^4)\ u + (t + t^3)\sin(x)\sin(y), & (x, y, t) \in \Omega_{\mathbf{x}} \times (0, 1], \\ u(x, y, 0) = \sin(x)\sin(y), \text{ on } \overline{\Omega}_{\mathbf{x}}, \\ u(x, y, t) = 0, \quad \text{on } \partial\Omega_{\mathbf{x}} \times [0, T]. \end{cases}$$

The exact solution of Example 1 is unknown. We obtain the series terms of the ADM algorithm, explained in Section 3.1. Using (6) with $L = D_t^1$ and initial condition, we obtain

$$u_0 = \sin(x)\sin(y) + L^{-1} f(x, y, t),$$

$$u_0 = \left(\frac{t^4}{4} + \frac{t^2}{2} + 1\right)\sin(x)\sin(y),$$

$$u_1 = -\left(\frac{2t}{5} + \frac{2t^3}{15} + \frac{2t^5}{25} + \frac{3t^7}{140} + \frac{t^9}{180}\right)\sin(x)\sin(y).$$

Similarly, we can obtain the other series terms u_n by the recurrence relation (6) and Φ_p for some integer p from (7). We produce the maximum residual error corresponding to ADM as follows

$$E_p^\infty = \max_{(x_i,y_j,t_n)\in\Omega^{m_x,m_y,n}} |L(\Phi_p(x_i, y_j, t_n)) + R(\Phi_p(x_i, y_j, t_n)) - f(x_i, y_j, t_n)|.$$

The effectiveness of the ADM method is shown in the Table 2, which clearly shows that the ADM-based approximate solution $\Phi_p(x, y, t)$ defined at (7) converges as p increases, at every time level. This matches with theoretical convergence results as explained in Section 4.1.

Now, we illustrate the effectiveness of the ADM-PINN algorithm over the ADM and PINN. Here, we provide a specific number of series terms, say $p = P$, and a uniform number of partitions to ADM-PINN. Table 3 clearly indicates that the proposed ADM-PINN provides better results than ADM and PINN.

In the Table 4, we have observed that as the number of series terms increases in the ADM-PINN algorithm, it significantly reduces the number of epochs required to satisfy the stopping criterion. We have also observed that increasing the number of partitions within a fixed series term of the ADM-PINN algorithm leads to a higher number of training epochs required to satisfy the stopping criteria. Surface plot and loss versus error relation are highlighted in the Fig. 2.

Table 2
Absolute maximum errors at each time level for Example 1.

Time (t)	$p = 7$	$p = 8$	$p = 9$	$p = 10$
0.1	2.3105E-12	1.3224E-14	6.6233E-17	2.9485E-19
0.2	1.5497E-10	1.7829E-12	1.7948E-14	1.6062E-16
0.3	1.9151E-09	3.3338E-11	5.0784E-13	6.8770E-15
0.4	1.2144E-08	2.8550E-10	5.8741E-12	1.0744E-13
0.5	5.4708E-08	1.6362E-09	4.2832E-11	9.9688E-13
0.6	2.0315E-07	7.4610E-09	2.3988E-10	6.8578E-12
0.7	6.7493E-07	2.9779E-08	1.1504E-09	3.9527E-11
0.8	2.1096E-06	1.1030E-07	5.0517E-09	2.0578E-10
0.9	6.4100E-06	3.9395E-07	2.1215E-08	1.0164E-09
1.0	1.9344E-05	1.3913E-06	8.7726E-08	4.9226E-09

Table 3
L_∞ error corresponding to series solution up to $p = P$ for Example 1.

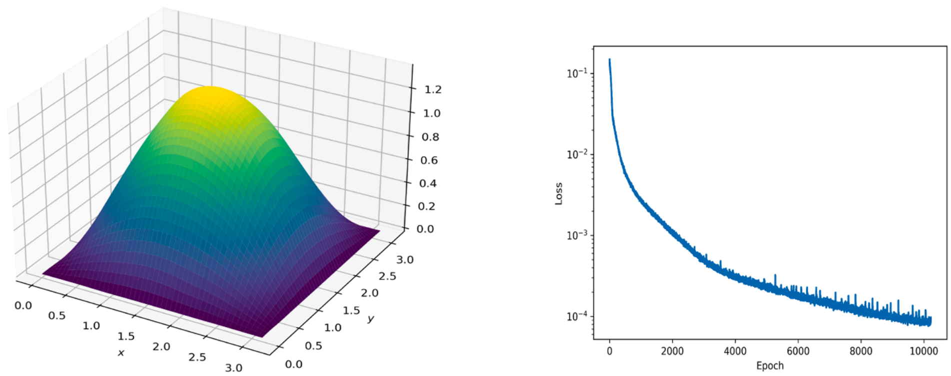
Series terms	Partition	ADM	PINN	fPINN	ADM-PINN
P = 1	N = 10	1.4000	6.5551E-2	9.7805E-2	9.7030E-2
	N = 15	1.3847	6.3747E-2	7.7214E-2	7.5248E-2
	N = 20	1.4000	4.8827E-2	6.9507E-2	6.1597E-2
	N = 25	1.3944	3.7402E-2	4.6110E-2	4.1236E-2
P = 2	N = 10	5.1225E-1	6.5551E-2	9.7805E-2	7.9433E-2
	N = 15	5.0665E-1	6.3747E-2	7.7214E-2	6.2549E-2
	N = 20	5.1225E-1	4.8827E-2	6.9507E-2	4.0023E-2
	N = 25	5.1023E-1	3.7402E-2	4.6110E-2	3.2894E-2
P = 3	N = 10	1.1650E-1	6.5551E-2	9.7805E-2	4.7064E-2
	N = 15	1.1523E-1	6.3747E-2	7.7214E-2	5.5982E-2
	N = 20	1.1650E-1	4.8827E-2	6.9507E-2	3.1738E-2
	N = 25	1.1604E-1	3.7402E-2	4.6110E-2	1.1151E-2

Table 4
Relation between epoch and series terms through Algorithm 2 for Example 1, when $N = 20$ number of uniform partitions are provided.

Series terms	P = 1	P = 2	P = 3
Epochs	11,325	10,241	9775
Time (sec.)	226.33	189.86	181.90

We know that different components of the loss function can have different convergence rates. Therefore, appropriate parameter tuning in the loss function is important to control the dominance of the single loss component. From ADM theory, it is clear that as the number of series terms increases in the ADM solution, max error decreases. Therefore, ADM initially may start with a large max error, and thereafter the error decreases as series terms increase. In Example 1, when $P = 1$, the max error is very high, and if we provide a large value for λ_4 in the ADM-PINN loss, the ADM-PINN algorithm learns the wrong direction, as mentioned in Fig. 3. Hence, we initially provide less weight to the ADM loss, and reduce the dominance of the ADM loss. Thereafter, we increase the value of λ_4 as the series terms increase, which increases the dominance of the ADM loss. This procedure provides a better direction for optimal neural parameters. We have also observed that fPINN requires 205.33 sec. computation time to achieve the $10k$ epochs, where ADM-PINN for $P = 1, 2, 3$ requires $201.84, 191.39, 186.08$ sec. computation time, respectively. We have also highlighted the impact on the overall convergence based on the tuning parameter λ_4, in Fig. 4.

Note that the initial data in Example 1 satisfies the infinite order compatibility conditions. Now we produce one example on a bounded domain which shows that the Adomian method can produce the exact solution too. In addition the following example shows that the sufficient conditions obtained in Section 4.2.1 are not necessary for the convergence of the Adomian approach.



(a) Surface plot at $t = 1$ (b) Loss versus epoch error

Fig. 2. Performance of present ADM-PINN algorithm for Example 1 at $t = 1$, when $M_x = 20, M_y = 20, N = 20$ numbers of uniform partitions and $P = 2$ number of partial terms in the partial sum are taken.

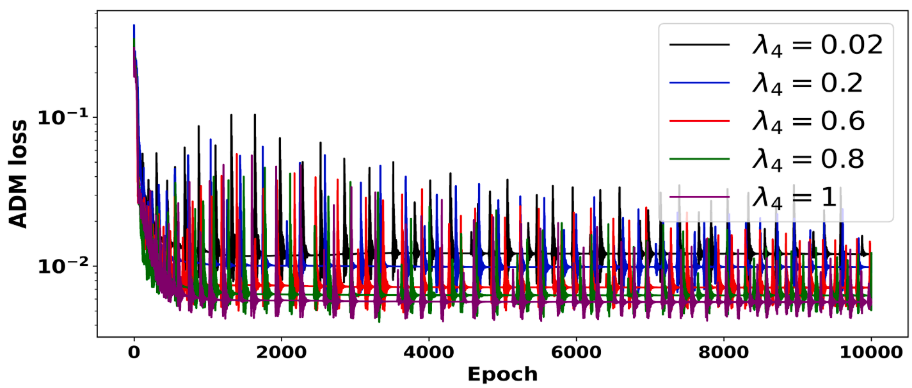


Fig. 3. Hyperparameter sensitivity analysis of λ_4 for Example 1, when $P = 1$.

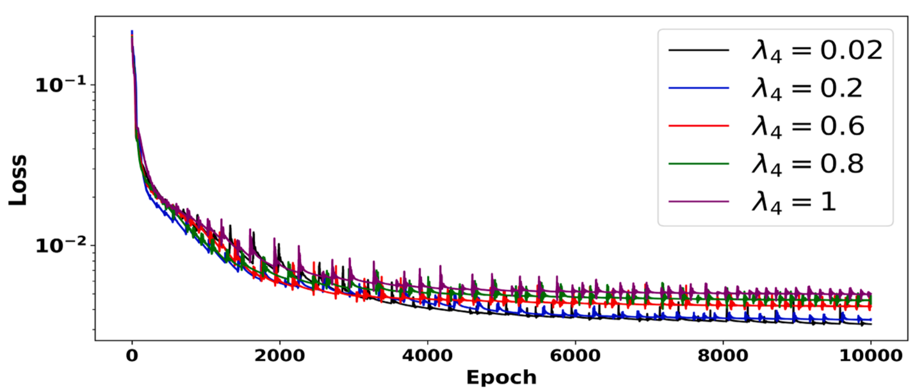


Fig. 4. Impact on the convergence of ADM-PINN based on the different values of the hyperparameter λ_4 for Example 1, when $P = 1$.

Example 2. Now we consider the following time fractional two-dimensional reaction-dominated model on $\Omega_{\mathbf{x}} = \Omega_x \times \Omega_y = (0, \frac{3}{2}) \times (0, \frac{3}{2})$:

$$\begin{cases} D_t^{\frac{4}{5}} u = 2\Delta u - (xyt + 1)\, u + f(x, y, t), \quad (x, y, t) \in \Omega_{\mathbf{x}} \times (0, 1], \\ u(x, y, 0) = xy(3 - 2x)(3 - 2y), \text{ on } \overline{\Omega}_{\mathbf{x}}, \\ u(x, y, t) = 0 \quad \text{on} \quad \partial\Omega_{\mathbf{x}} \times [0, T], \end{cases}$$

where $f(x, y, t) = (\frac{12}{\Gamma(16/5)} t^{11/5} + \frac{\Gamma(5/3)}{\Gamma(13/5)} t^{-2/15}) xy(3 - 2x)(3 - 2y) + 8(1 + t^{(2/3)} + 2t^3)(3x + 3y - 2x^2 - 2y^2) + (1 + xyt)(1 + t^{(2/3)} + 2t^3) xy(3 - 2x)(3 - 2y)$. For this example, compatibility conditions hold up to first order. Let us decompose f by $f(x) = f_1(x) + f_2(x)$, where $f_1(x) = (\frac{12}{\Gamma(16/5)} t^{11/5} + \frac{\Gamma(5/3)}{\Gamma(13/5)} t^{-2/15}) xy(3 - 2x)(3 - 2y)$ and $f_2(x) = 8(1 + t^{(2/3)} + 2t^3)(3x + 3y - 2x^2 - 2y^2) + (1 + xyt)(1 + t^{(2/3)} + 2t^3) xy(3 - 2x)(3 - 2y)$.

Now, the relation (6) with $L = D_t^{\frac{4}{5}}$ and the initial condition in Example 2 follows that

$$u_0 = xy(3 - 2x)(3 - 2y) + L^{-1} f_1(x, y, t),$$

which implies $u_0 = (1 + t^{(2/3)} + 2t^3) xy(3 - 2x)(3 - 2y)$. Again from (6), we have

$$u_1 = L^{-1} f_2(x, y, t) + L^{-1}(2\Delta u_0 - (xyt + 1)u_0)),$$

which leads to $u_1(x, y, t) = 0$. Therefore, the next terms of the recurrence relation will become zero. Hence, we obtain the $u(x, y, t) = u_0(x, y, t) =$

Table 5
Absolute maximum errors for Example 3.

Time (t)	$p=7$	$p=8$	$p=9$	$p=10$
0.1	6.58831E-11	1.22986E-12	2.12506E-14	3.44529E-16
0.2	1.67475E-09	4.96187E-11	1.36230E-12	3.49344E-14
0.3	1.11262E-08	4.31831E-10	1.55330E-11	5.21856E-13
0.4	4.26894E-08	2.00632E-09	8.73983E-11	3.55622E-12
0.5	1.21283E-07	6.61064E-09	3.34023E-10	1.57665E-11
0.6	2.85016E-07	1.75304E-08	9.99740E-10	5.32675E-11
0.7	5.87739E-07	4.00278E-08	2.52820E-09	1.49214E-10
0.8	1.10171E-06	8.19323E-08	5.65244E-09	3.64458E-10
0.9	1.92045E-06	1.54300E-07	1.15044E-08	8.01839E-10
1.0	3.16187E-06	2.72150E-07	2.17454E-08	1.62467E-09

$(1+t^{(2/3)}+2t^3)xy(3-2x)(3-2y)$ which satisfies the above boundary conditions. Hence, ADM converges to the original solution in one iteration.

Now we consider the following two dimensional time fractional Schrodinger equation (Lashkarian & Hejazi, 2018) to show the effectiveness of the present approach.

Example 3. Consider the following time fractional Schrodinger equation in 2D on $\Omega_{\mathbf{x}}=\Omega_x\times\Omega_y=(-\frac{\pi}{4},\frac{\pi}{4})\times(-\frac{\pi}{4},\frac{\pi}{4})$:

$$\begin{cases} D_t^{2/3}u = i(a\ \Delta u - b\ u)+f(x,y,t), \quad (x,y,t)\in\Omega_{\mathbf{x}}\times(0,1],\\ u(x,y,0)=2\cos(2x)\cos(2y),\ \text{on}\quad \overline{\Omega}_{\mathbf{x}},\\ u(x,y,t)=0, \quad \text{on} \quad \partial\Omega_{\mathbf{x}}\times[0,T]. \end{cases}$$

In the above example, we take $a=0.024$, $b(t)=0.078125$ and $f(x,y,t)=\left(\frac{1}{\Gamma[7/3]}t^{4/3}\left(2+\frac{27}{14}t\right)+i\frac{2161}{4000}\left(1+\frac{1}{2}t^2+\frac{3}{8}t^3\right)\right)\cos(2x)\cos(2y)$, respectively. Here $i=\sqrt{-1}$ is a complex number. For this choice of f, the exact solution of Example 3 is $u(x,y,t)=(2+t^2+\frac{3}{4}t^3)\cos(2x)\cos(2y)$. Now let us find the approximate solution based on ADM. Using (6) with $L=D_t^{\frac{2}{3}}$ and the initial condition, we obtain

$$u_0=2\cos(2x)\cos(2y)+L^{-1}f(x,y,t),$$

which implies

$$u_0(x,y,t)=\left(2+\frac{2161i}{4000\Gamma[5/3]}t^{2/3}+t^2+\frac{2161i}{4000\Gamma[11/3]}t^{8/3}\right.$$
$$\left.+\frac{3}{4}t^3+\frac{1949i}{16000\Gamma[14/3]}t^{11/3}\right)\cos(2x)\cos(2y).$$

The other terms of the partial sum can be obtained in a similar manner by the relation (6). Now, we produce the maximum absolute error in favor of our proposed approach. As the exact solution is known, the maximum absolute error over the discrete domain is defined as follows:

$$E_p^{\infty}=\max_{(x_i,y_j,t_n)\in\Omega^{m_x,m_y,n}}\left(\text{Abs}\Big(u(x_i,y_j,t_n)-\Phi_p(x_i,y_j,t_n)\Big)\right),$$

Here $\Phi_p(x,y,t)$ is defined at (7). Table 5 produces the maximum error for ADM solution at a fixed time level for different values of p. This table clearly shows that the approximate solution converges for larger values of p.

Here, we aim to calculate the Algorithm 2 based solution and compare it with the ADM and fractional PINN based solutions. From Table 6, it is very clear that the proposed algorithm performs better than the existing ADM and fractional PINN. It also observes that as the number of training points increases in the proposed algorithm, the maximum error of the proposed algorithm decreases.

Table 6
L_∞ error corresponding to series solution up to $p=P$ for Example 3.

Series terms	Partition	ADM	PINN	fPINN	ADM-PINN
P = 1	N = 10	8.1573E-1	–	7.1258E-1	2.5461E-1
	N = 15	8.0681E-1	–	6.8569E-1	2.1864E-1
	N = 20	8.1573E-1	–	6.5723E-1	1.9972E-1
	N = 25	8.1251E-1	–	6.4715E-1	1.1943E-1
P = 2	N = 10	1.4651E-1	–	7.1258E-1	3.0271E-2
	N = 15	1.4490E-1	–	6.8569E-1	2.1473E-2
	N = 20	1.4651E-1	–	6.5723E-1	1.8021E-2
	N = 25	1.4593E-1	–	6.4715E-1	1.0194E-2
P = 3	N = 10	2.2092E-2	–	7.1258E-1	3.1468E-3
	N = 15	2.1850E-2	–	6.8569E-1	2.4367E-3
	N = 20	2.2092E-2	–	6.5723E-1	1.7132E-3
	N = 25	2.2004E-2	–	6.4715E-1	1.0011E-3

Table 7
L_∞ error corresponding to series solution up to $p=P$ for Example 4.

Series terms	Partition	ADM	PINN	fPINN	ADM-PINN
P = 1	N = 10	5.4887E-1	–	9.7805E-2	4.3152E-2
	N = 15	5.4887E-1	–	7.7214E-2	3.9705E-2
	N = 20	5.4887E-1	–	6.9507E-2	2.8371E-2
	N = 25	5.4887E-1	–	4.6110E-2	1.4824E-2
P = 2	N = 10	3.0270E-1	–	9.7805E-2	3.6187E-2
	N = 15	2.9789E-1	–	7.7214E-2	2.3969E-2
	N = 20	3.0270E-1	–	6.9507E-2	2.0156E-2
	N = 25	3.0104E-1	–	4.6110E-2	1.2843E-2
P = 3	N = 10	1.7327E-1	–	9.7805E-2	3.1687E-2
	N = 15	1.7327E-1	–	7.7214E-2	2.1486E-2
	N = 20	1.7327E-1	–	6.9507E-2	1.8963E-2
	N = 25	1.7327E-1	–	4.6110E-2	9.9469E-3

Example 4. Consider the following two dimensional time fractional problem on $\Omega_{\mathbf{x}}=(0,\pi)\times(0,\pi)$:

$$\begin{cases} D_t^{1/2}u = a\ \Delta u - b(x,y,t)\ u+f(x,y,t), \quad (\mathbf{x},t)\in\Omega_{\mathbf{x}}\times(0,1],\\ u(x,y,0)=0,\ \text{on}\quad \overline{\Omega}_{\mathbf{x}},\\ u(x,y,t)=0, \quad \text{on} \quad \partial\Omega_{\mathbf{x}}\times[0,T]. \end{cases}$$

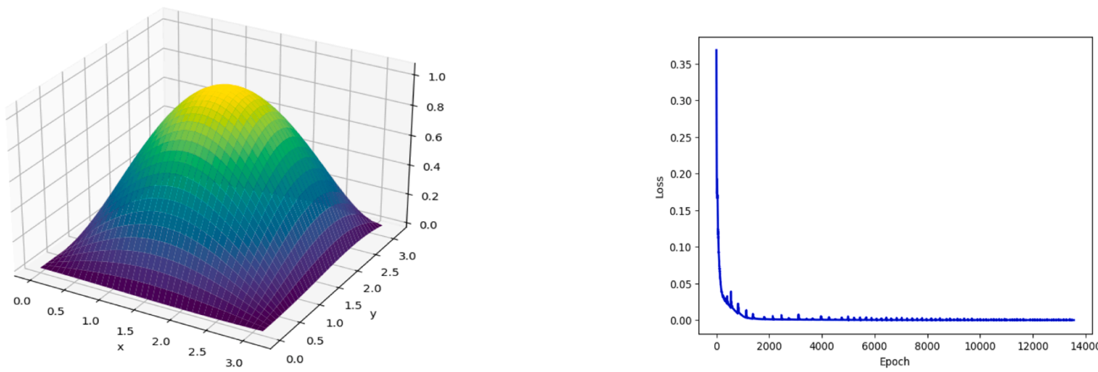
In the above example, we take $a=0.1$, $b(x,y,t)=\frac{1}{5}(xyt^2)$. The source function $f(x,y,t)$ is calculated in such a way that the exact solution of Example 4 satisfies $u(x,y,t)=t^{1/4}\sin(x)\sin(y)$. From the exact solution, it is clear that the function has a singularity in time. Now, we use the defined relations (6) to find the series terms of the ADM-based approximated solution as follows

$$u_0(x,y,t)=\left(t^{(1/4)}+\frac{3\pi t^{(3/4)}}{40\sqrt{2}(\Gamma(7/4))^2}+\frac{27\pi t^{(11/4)}xy}{616\sqrt{2}(\Gamma(7/4))^2}\right)\sin x\sin y.$$

The other terms of the partial sum can be obtained similarly, through the defined relations (6). Note that this example does not satisfy the infinite order compatibility condition. However, the experimental results in the tables show the effectiveness of the proposed PINN algorithm over the ADM and PINN. Table 7 clearly indicates the convergence of the ADM numerically, when the number of terms in the partial sum p increases. It also indicates that the proposed algorithm provides better results than ADM and PINN as the number of series terms increases. Table 7 also illustrates the convergence behavior of the proposed algorithm as the number of uniform partitions increases; the error of the proposed algorithm decreases.

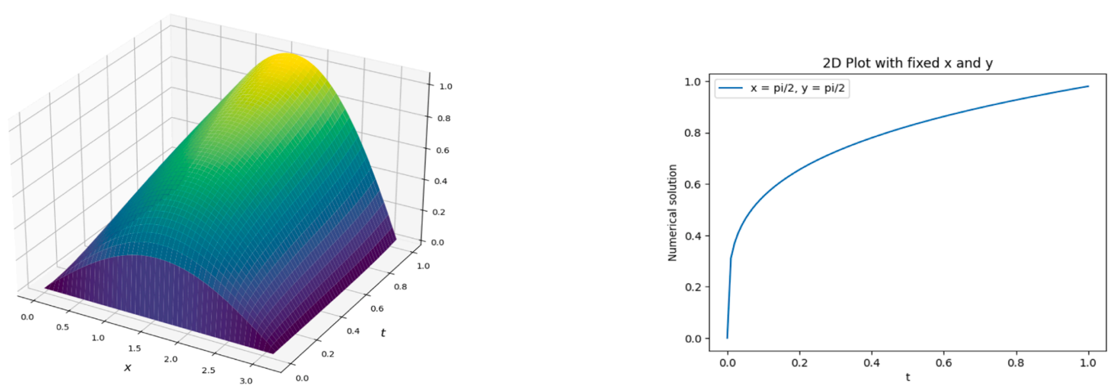
We have also highlighted that as the number of series terms increases, the number of epochs required to reach the stopping criteria decreases in Table 8.

From the proposed algorithm, we provide $N=20$, $M_x=20$, $M_y=20$ numbers of uniform partitions and number of series terms up to $P=2$ to show the surface plot and the loss versus error relation shown from the proposed algorithm is mentioned in the Fig. 5.



(a) Surface plots at fix $t = 1$ (b) performance of loss versus epoch

Fig. 5. Performance of the proposed algorithm for Example 4.



(a) $y = \frac{\pi}{2}$ (b) Cross section at $x = \frac{\pi}{2}$, $y = \frac{\pi}{2}$.

Fig. 6. Layer solution at fix $y = \pi/2$, and cross section at fix $x = \pi/2, y = \pi/2$ using proposed algorithm for Example 4.

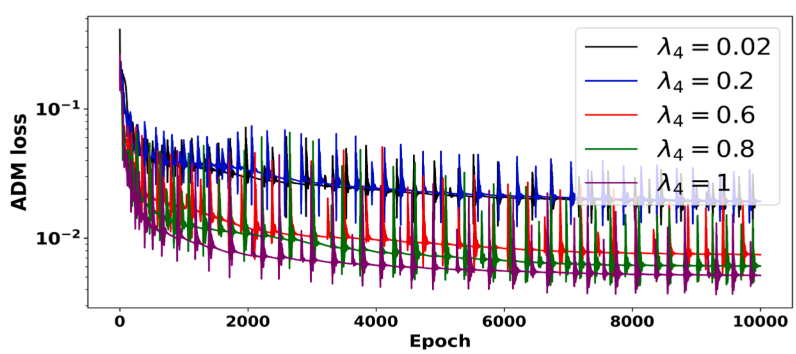


Fig. 7. Hyperparameter sensitivity analysis of λ_4 for Example 4, when $P = 1$.

Table 8
Relation between epoch and number of terms in series in Algorithm 2 for Example 4, when $N = 20$ number of uniform partitions are provided, and tolerance is 5×10^{-5}.

Series terms	P = 1	P = 2	P = 3
Epochs	32,861	15,269	13,558
Time (sec.)	2023.59	940.27	418.46

Figs. 6a and 6b indicate the behavior of the initial time layer for the time fractional reaction diffusion problem of Example 4 over the space-time domain, by fixing y direction, and also shows the cross-sectional view of surface plot at $x = \pi/2, y = \pi/2$, which indicates that the initial time layer has been captured efficiently through the proposed algorithm.

From the theoretical convergence of ADM, it is clear that as the series terms increase, it converges to the exact solution. Hence, the ADM solution may have a large error at the initial series terms. Therefore, the high weight at initial series terms to the ADM loss may affect the convergence of the ADM-PINN loss, which is highlighted in Figs. 7 and 8.

We employed a non-uniform/adaptive mesh-based PINN framework, using a graded mesh to provide a high concentration of mesh points in regions where the solution is non-smooth. The spatial points are generated uniformly over the computational domain, and temporal points are generated using $t_n = T(n/N_t)^r, \quad n = 1, 2, \cdots, N_t$, where T is the final

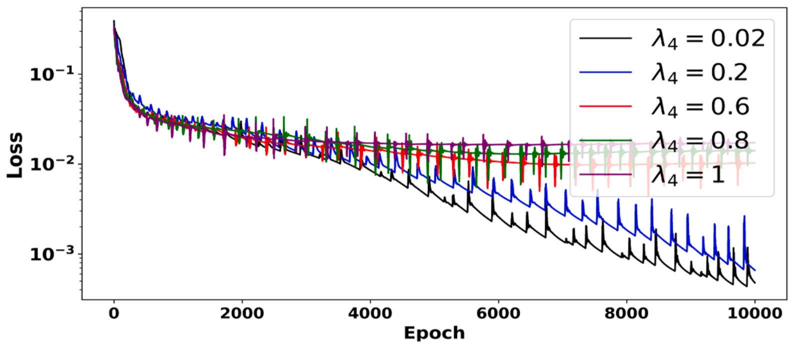


Fig. 8. Impact on the convergence of ADM-PINN based on the different values of hyperparameter λ_4 for Example 4, when $P = 1$.

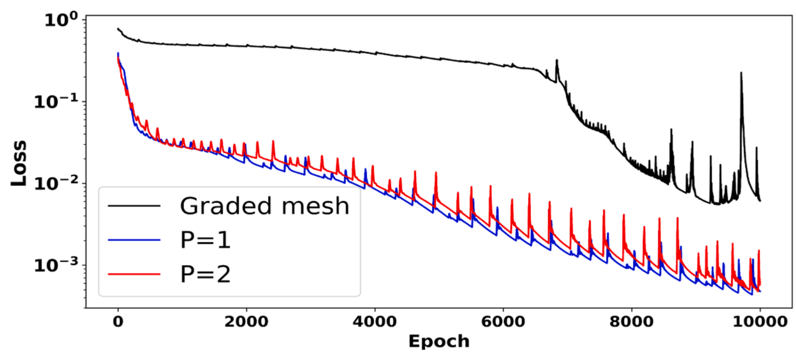


Fig. 9. Comparison of ADM-PINN loss, for $P = 1,\ 2$ and, the time-graded mesh based PINN loss for Example 4.

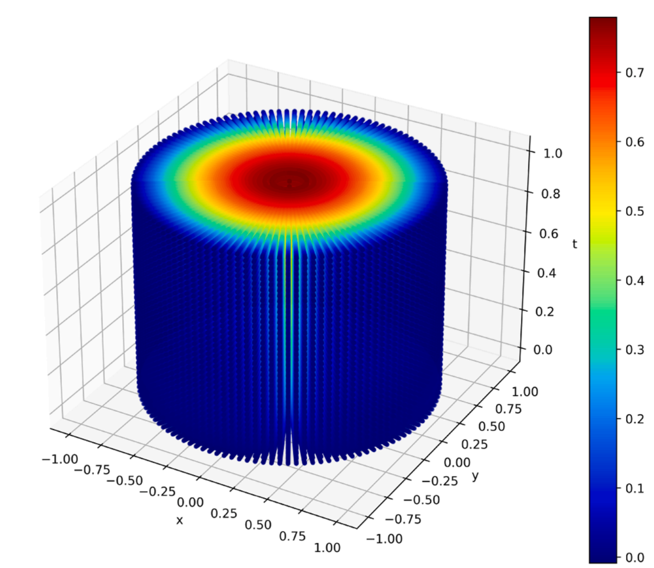


Fig. 10. Approximate solution on the circular domain along with the time domain for Example 5.

time, N_t denotes the total number of points in the temporal domain, and r denotes the grading parameter. The grading parameter is computed as $r = (2 - \alpha)/\alpha$, where α is a fractional order parameter. Then, the complete mesh is generated from the given spatial and temporal points. A graded mesh is employed in the fPINN framework and compared with the ADM-PINN algorithm for Example 4. The hyperparameter λ_4 is set to 0.02 for both $P = 1$ and $P = 2$. The comparison is highlighted in Fig. 9.

Example 5. Now, we check the effectiveness of the ADM and ADM-PINN algorithm for the time fractional reaction-diffusion problem on the circular domain, which is 2D cross-section of a pipe. The set of points inside the circular domain is defined as $\Omega = \Omega_{\mathbf{x}} \times (0, T]$ whose boundary is $\partial\Omega = \partial\Omega_{\mathbf{x}} \times [0, T]$, where $\Omega_{\mathbf{x}} = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < R^2\}$, $\partial\Omega_{\mathbf{x}} = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = R^2\}$, and R denotes the radius of the circle. We take $\alpha = \frac{1}{2}$, $a = \frac{1}{20}(1 - x^2 - y^2)$, $b = \frac{t}{10}$, $f = t^{1/3}(1 - x^2 - y^2)$, $R = 1$, $T = 1$ and initial condition $s(\mathbf{x}) = 0$ in the defined Model (1).

Now, we use the recurrence relation (6) to find the series terms of the ADM approximation. The first term of the ADM approximation is defined as follows:

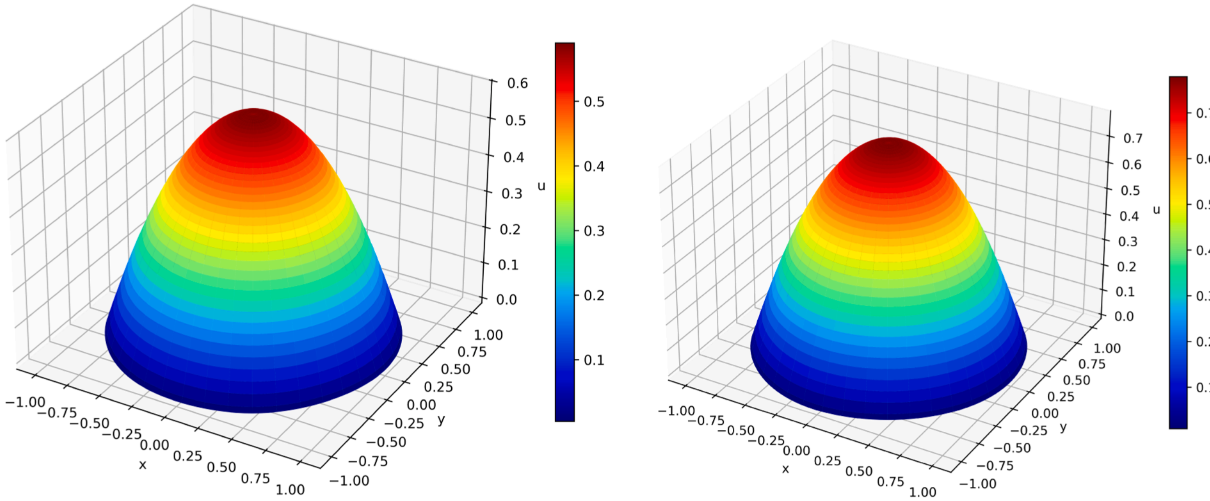
$$u_0(x, y, t) = \frac{\Gamma\left(\frac{4}{3}\right)}{\Gamma\left(\frac{11}{6}\right)} t^{5/6}\left(1 - x^2 - y^2\right).$$

Similarly, the other series terms can be calculated through the defined recurrence relation (6). We mainly highlight a few terms as follows:

$$u_1(x, y, t) = \left(-\frac{3t^{4/3}}{10} - \frac{33t^{7/3}}{1120}\right)\left(1 - x^2 - y^2\right),$$

$$u_2(x, y, t) = \frac{\Gamma\left(\frac{7}{3}\right)}{\Gamma\left(\frac{17}{6}\right)}\left(\frac{3t^{11/6}}{25} + \frac{3t^{17/6}}{136} + \frac{33t^{23/6}}{31280}\right)\left(1 - x^2 - y^2\right).$$

From the above series terms, it is clear that the ADM approximation satisfies the initial and boundary conditions for the circular domain. The experiments are performed on a workstation equipped with 256 GB system RAM and a 16 GB GPU memory. During execution, the peak memory utilization was approximately 4% of the system RAM (almost 10.2 GB) and 65% of the GPU memory (almost 10.4 GB). We took 12000 interior data, 4000 boundary data, and 800 initial condition data to train the model on a circular domain and demonstrate the approximation of the ADM-PINN solution at fixed time levels $t = 0.5,\ 1$ in Fig. 11a and b. Table 9 clearly indicates the convergence behavior of the ADM algorithm as the number of series terms increases. Now, we utilize these partial terms in the ADM-PINN algorithm to check the effectiveness of the proposed algorithm on the circular domain. The complete approximation along the time level is highlighted in Fig. 10, where different



(a) $t = 0.5$ (b) $t = 1$

Fig. 11. Approximate solution at fix t for Example 5.

Table 9
L_∞ error corresponding to series solution up to $p = P$ for Example 5.

Series terms	$P = 1$	$P = 2$	$P = 3$
ADM	4.2719E-1 (26.19sec.)	1.4825E-1 (134.91sec.)	4.4463E-2 (308.84sec.)
fPINN	9.6090E-2 (298.73sec.)	9.6090E-2 (298.73sec.)	9.6090E-2 (298.73sec.)
ADM-PINN	7.7034E-2 (290.63sec.)	5.5283E-2 (280.98sec.)	1.2919E-2 (248.81sec.)

colors indicate the approximate value of the ADM-PINN solution. Note that ADM-PINN takes relatively smaller computational time compare to ADM solution in order to keep the same level of accuracy as depicted in Table 9. Hence, the effectiveness of the ADM-PINN algorithm is not limited to the bounded rectangular domain.

7. Conclusions

We have successfully developed an ADM-PINN based deep learning algorithm for the time fractional reaction diffusion problems in 2D. The theoretical bounds based on semi discrete scheme in time and PINN approximation in space is also derived and the convergence of the ADM-PINN for the time fractional reaction diffusion initial boundary value problems is established. First, we derive the convergence of the Adomian method and few sufficient conditions on the data under which the Adomian decomposition solution based on the initial condition will converge to the exact solution of the model. In addition, we explain the procedure of obtaining the numerical approximation based on ADM and ADM-PINN. Further, we performed numerical experiments to verify that the ADM-PINN based deep learning algorithm performs better than ADM and PINN in terms of error.

CRediT authorship contribution statement

Arihant Patawari: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Pratibhamoy Das:** Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization; **Subrata Rana:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Ethics approval

All authors provide the consent for submission.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
In addition they declare that this is not under consideration to anywhere else. The authors are thankful to the anonymous referees for their valuable comments in order to improve this work.

Data availability

Data sharing not applicable for this work as no data is generated.

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